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Experimental characterization and hyperelastic constitutive modeling of open-cell elastomeric foams

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ABSTRACT

Open-cell elastomeric foams – materials consisting of an elastomeric matrix and a connected pore space – exhibit mechanical behavior marked by high compressibility and strong coupling between the volumetric and distortional responses. In this paper, we present a methodology for the experimental characterization and constitutive modeling of non-localizing, isotropic, open-cell elastomeric foam materials under quasi-static, equilibrium loading. We conduct large-deformation, homogeneous simple compression/tension experiments on three relative densities of a polyurethane-based elastomeric foam to inform a phenomenological, isotropic, hyperelastic constitutive model. The model is based on the invariants of the logarithmic strain and accounts for high compressibility and strong volumetric-distortional coupling. To validate the predictive capability of the model, we consider three types of validation experiments that involve inhomogeneous deformation: spherical and conical indentation, simple-shear-like deformation both without and with a fixed amount of pre-compression, and tension of a specimen with circular holes. We compare load-displacement responses as well as full displacement fields from the validation experiments against corresponding model predictions obtained using finite-element-based numerical simulations and demonstrate that the model is capable of accurately capturing the experimental response.

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1. Introduction

Open-cell elastomeric foams are materials consisting of two continuous phases: a polymeric elastomer and a connected gas-filled pore space. Due to the elastomeric matrix and porous structure, these materials are highly compliant and capable of undergoing large, reversible deformations involving substantial volume change. Elastomeric foams are widely used in situations in which compliance and compressibility are necessary, such as in impact protection and cushioning, and appear in a wide variety of applications.

The mechanical behavior of open-cell elastomeric foams is quite complex, involving large deformations, substantial compressibility, and strong tension/compression asymmetry. Consider, as an illustrative example, the behavior of a commercially-available impact protection foam – a polyurethane-based, open-cell elastomeric foam that will be the focus of our work – whose response to external load is demonstrated in Fig. 1 for three different relative densities (Poron XRD, Rogers Corp, Rogers, CT). For each density, Fig. 1 shows the experimentally-measured engineering stress versus axial engineering strain

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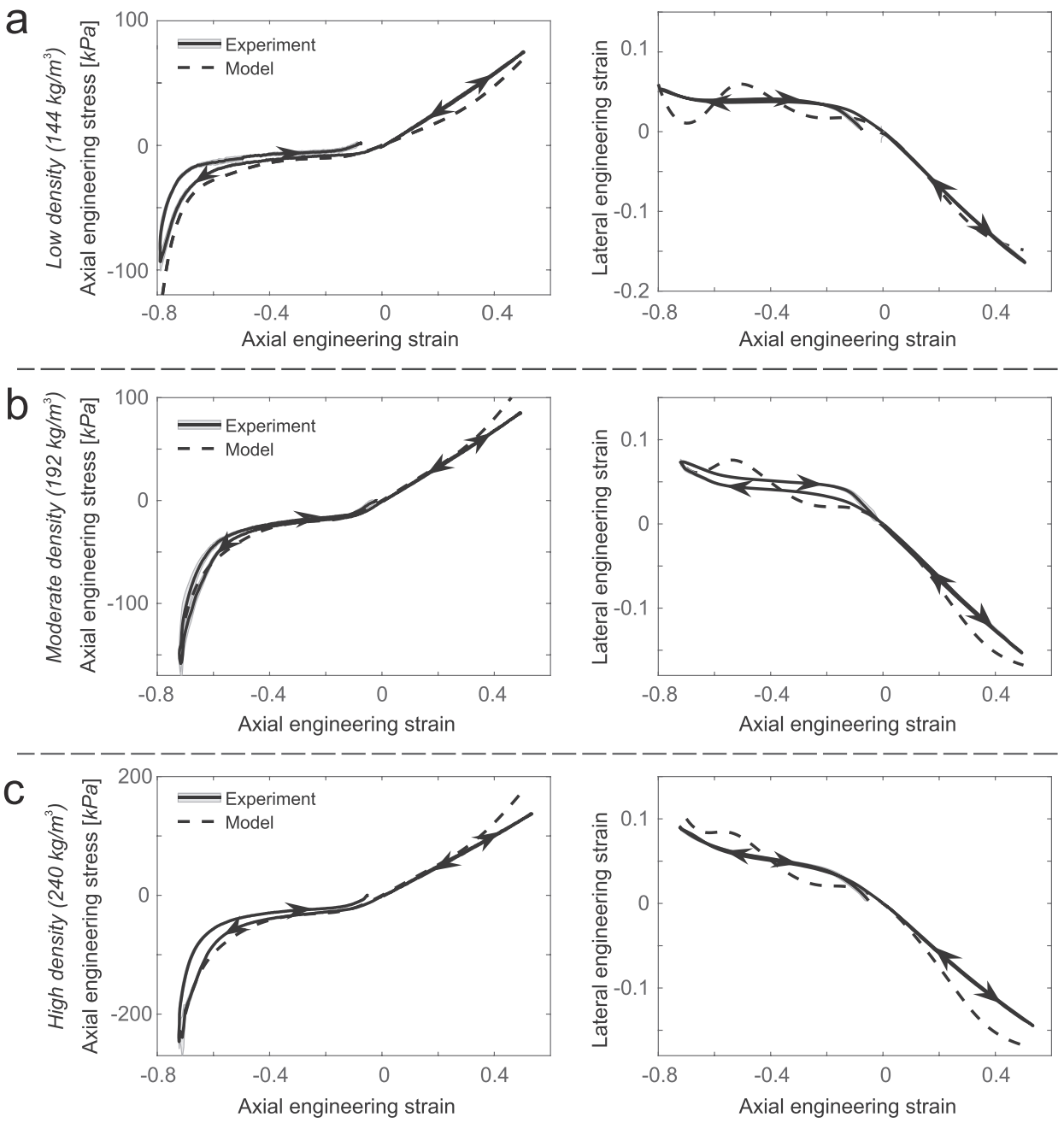


Fig. 1. (Left) Axial engineering stress versus axial engineering strain curves and (right) lateral engineering strain versus axial engineering strain curves for polyurethane-based, open-cell elastomeric foams of (a) low (144 kg/m^3), (b) moderate (192 kg/m^3), and (c) high (240 kg/m^3) density with the same matrix composition. The solid lines with shaded error areas (one standard deviation, nine experiments) indicate experimental data, and the dashed lines are constitutive model fits. Note that for (a), (b), and (c), the stress axes differ in range. For all densities, note both the plateau regime in compression in both the stress and lateral strain histories as a function of axial strain, and the small lateral strain magnitudes under large axial strain magnitudes as characteristic features of deformation in open-cell elastomeric foams.

response in the left column and the corresponding lateral engineering strain versus axial engineering strain response in the right column for quasi-static (low strain-rate) simple compression/tension (experimental details are discussed in Section 2). The response in all three cases is highly nonlinear and displays substantial asymmetry between tension and compression. In particular, in compression, the response is initially linear, but as the axial strain magnitude increases, the response gradually transitions into a “plateau” in which both the axial stress and the lateral strain vary minimally as the axial strain magnitude increases. The plateau regime is generally associated with buckling in the elastomeric-matrix microstructure and concomitant closure of the pore space (Gibson and Ashby, 1997). Importantly, the plateau regime for Poron XRD is not

completely flat, and we observe homogeneous deformation with no evidence of strain localization in our compression experiments. Then, as the material is compressed beyond the plateau regime, rapid stiffening is observed in the axial stress response. However, in tension neither a plateau regime nor a rapidly stiffening response is observed. Upon unloading in both compression and tension, some hysteresis but no permanent set is observed. This is in contrast to the inelastic behavior of crushable foams (e.g., Deshpande and Fleck, 2001; Liu and Subhash, 2004; Neilsen et al., 1995; Zhang et al., 1997), which yield and display permanent set upon unloading. In this work, we focus on the mechanics of elastic foams that exhibit neither deformation localization nor permanent set.

Due to the complexity of the mechanical behavior of open-cell elastomeric foams, developing predictive constitutive models remains challenging despite the ubiquity of these materials. There is a substantial history of constitutive modeling of porous and foam materials (e.g., Gibson and Ashby, 1997; Gong et al., 2005). Here, we focus on hyperelastic modeling approaches for capturing the equilibrium, elastic response described in the preceding paragraph. Commonly, Ogden-type hyperelastic models (Ogden, 1972a; Storakers, 1986), which utilize a phenomenological free-energy density function that depends on the principal stretches, are used to capture data for highly-compressible foams in the large-stretch regime (e.g., Briody et al., 2012; Chen et al., 2012; Jemiolo and Turteltaub, 2000; Ju et al., 2013; Mills and Gilchrist, 2000). While models of this type are capable of describing the features of the equilibrium, hyperelastic response of open-cell elastomeric foams, they can be unwieldy to work with (e.g., Petre et al., 2006; Widdle Jr. et al., 2008). Material parameter fitting is a challenge, and a fitted model typically has minimal extrapolative potential, often providing poor predictions in loading scenarios not used in fitting or becoming unstable beyond the fitted range. Other hyperelasticity models utilize free-energy density functions that depend on the principal invariants of the Cauchy-Green deformation tensors, including the well-known Blatz-Ko hyperelasticity model (Blatz and Ko, 1962) and its extensions (e.g., Ciambella and Saccomandi, 2014; Murphy, 2000). The Blatz-Ko model can capture the response of compressible foams up to moderate stretches but is not equipped to describe large-stretch behavior – specifically, the plateau regime in compression. Other invariant-based approaches include polynomial-type free-energy density functions (Anani and Alizadeh, 2011; Yang and Shim, 2004). Furthermore, homogenization-based approaches (Danielsson et al., 2004; Lopez-Pamies and Ponte Castañeda, 2007a; 2007b; Shrimali et al., 2019) consider porous elastomers with microstructures consisting of closed-cell pores and result in explicit, invariant-based hyperelastic models that incorporate the effects of microstructural features, such as the void volume fraction, pore size dispersion, and pore shape. However, since these models invoke closed-cell microstructures in their derivations and do not account for buckling in elastomeric-matrix ligaments, they cannot be expected to capture the response of highly-compressible, open-cell foams under large compressive deformations – in particular, the response in the plateau regime. Finally, we note that fully-micromechanical models, based on strut-type unit cells (e.g., Brydon et al., 2005; Gong and Kyriakides, 2005; Sabuwala and Gioia, 2013; Shulmeister et al., 1998; Wang and Cuitiño, 2000) or unit cells involving cylindrical or spherical voids (e.g., Chen et al., 2018; Guo et al., 2008; Shrimali et al., 2019), have been used to capture the precise behavior of elastomeric foams with different microstructures. While such modeling approaches provide important physical insights, they are computationally expensive to apply to problems involving length scales much greater than the characteristic size of the porous microstructure, and hence, explicit hyperelastic models are valuable.

The purpose of this paper is to present a comprehensive methodology for the experimental characterization and constitutive modeling of non-localizing, isotropic, open-cell elastomeric foam materials under quasi-static, equilibrium loading. Initial model development and material parameter estimation are informed by the homogeneous simple compression/tension experiments shown in Fig. 1. Accurate full-field kinematic measurements in these experiments are obtained using the quality-factor-based digital image correlation (qDIC) procedure developed in our previous work (Landauer et al., 2018), which is crucial to measure both axial and lateral strains under large deformations. Rather than basing our isotropic hyperelasticity model on the principal invariants of the Cauchy-Green deformation tensors, our hyperelastic model adopts those of Criscione et al. (2000), which are based on the invariants of the logarithmic strain (Hencky, 1931; 1933). This choice is made on pragmatic grounds, since the logarithmic-strain invariants of Criscione et al. (2000) may be associated with specific aspects of deformation, enabling a more straightforward path to choosing and calibrating phenomenological fitting functions that capture experimental data. For perspective, a number of isotropic, logarithmic-strain-based hyperelasticity models have been proposed in the literature. Importantly, the work of Anand (1979, 1986), showed that a simple quadratic free-energy function employing the logarithmic strain is capable of capturing experimental data for a wide class of materials in the moderate-stretch regime. Since then, various models have been successfully applied to incompressible and nearly-incompressible materials in the large-stretch regime (e.g., Diani and Gilormini, 2005; Horgan and Murphy, 2009; Kakavas, 2000; Xiao, 2013). The present work extends this literature towards a constitutive modeling approach for highly-compressible materials. We decompose the free-energy function into terms that represent the coupled volumetric/distortional, purely distortional, and purely volumetric contributions to the response and select phenomenological fitting functions for each term that capture our experimental data for open-cell polyurethane foam. We then perform several validation tests that probe the predictive capability of the model in a variety of deformation modes, namely, compression, shear, and tension-dominated situations. Our intent is to provide an accessible characterization and modeling framework for highly-compressible elastic materials, and we expect that future works will consider other specific, perhaps micromechanically-based, fitting functions and extend the approach to other open-cell elastomeric foam materials.

The remainder of this paper is organized as follows. In Section 2, we describe our experimental program for characterizing the mechanical response of the isotropic, polyurethane-based elastomeric foam, marketed under the “Poron XRD” trademark, using homogeneous simple compression/tension experiments as well as our approach for validation testing. Then,

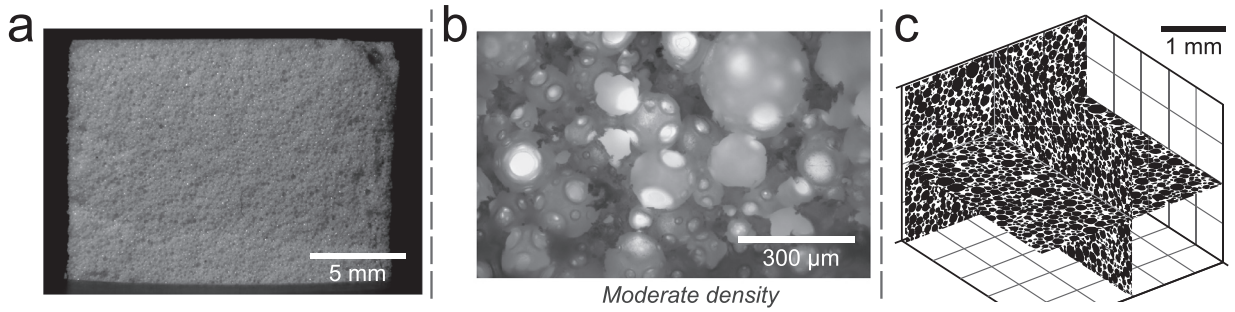


Fig. 2. Three images of a representative moderate density foam sample at various scales. (a) The cut face of a foam cuboid. (b) Optical microscopy image of the microstructure of the surface of a cut face. (c) Micro-computed tomography 3D volumetric render of the foam microstructure (watershed-based binarization, in which black is void).

in Section 3, we present our hyperelastic constitutive model for the elastic behavior of isotropic, open-cell elastomeric foams and apply the model to Poron XRD. In Section 4, the constitutive model is validated in three configurations: first, spherical and conical indentation, second, simple-shear-like deformation both without and with fixed-displacement pre-compression, and third, tension of a specimen with circular holes, where the conditions are specifically chosen to probe the model in compression, shear, and tension-dominated situations. In Section 5, we discuss issues of stability and compare the proposed model with other important models from the literature. Finally, we close with a summary and a discussion of the limits of the modeling approach and future research directions in Section 6.

2. Experimental program

In this section, we describe our process for experimentally testing the mechanical response of isotropic, elastomeric foams. We perform two classes of experiments: (1) homogeneous characterization experiments that are utilized to choose constitutive equations and estimate material parameters and (2) inhomogeneous validation experiments that are utilized to test constitutive model predictions. First, regarding characterization testing, we perform simple compression and simple tension experiments. All kinematic data in these experiments are obtained using full-field digital image correlation (DIC), which enables both axial and lateral strains to be obtained. Moreover, we are able to ensure that no unintended inhomogeneous deformation (e.g., specimen barreling or buckling, compaction banding, surface wrinkling, etc.) occurs during characterization experiments. Second, we perform a variety of inhomogeneous validation tests, namely, spherical and conical indentation, simple-shear-like deformation both without and with pre-compression, and tension of a specimen with circular holes, each of which are discussed further in Section 4. These validation experiments probe the predictive capability of the constitutive model in compression, shear, and tension-dominated situations.

Foam material. The experimental testing procedures and constitutive theory presented in this paper are intended to be applicable to a large class of isotropic, open-cell elastomeric foams, provided that the material is capable of deforming homogeneously and does not exhibit localized deformation, such as compaction banding. In the present work, we consider three different relative densities of a single type of foam material. Our selection is a polyurethane-based open-cell elastomeric foam developed by Rogers Corporation (Poron XRD, Rogers, CT) for impact mitigation applications, which has been the subject of previous studies in the literature (Tang et al., 2017; Yang and Shim, 2004). The three relative densities considered, as specified by the manufacturer, are 144 kg/m^3, 192 kg/m^3, and 240 kg/m^3, which correspond to void volume fractions within the approximate range 0.75–0.95 and are hereafter referred to as the low, moderate, and high density foams, respectively. To get a sense of the foam microstructure, Fig. 2 shows (a) a macro-scale image, (b) a micro-scale image, and (c) a 3D volume render obtained via micro-computed tomography (μCT) for a representative sample of the moderate density foam. The pore structure is highly random, and based on optical microscopy measurements, the mean pore diameter is approximately 70 μm with a maximum pore diameter of up to about 500 μm for the moderate density foam.

Specimen preparation. The details of specimen preparation depend on the specific type of test, but in general, specimens are collected from as-received approximately 12.9 mm-thick sheets of bulk material, speckled for digital image correlation, and then installed in appropriate fixtures. For compression testing, foam cuboids of approximately $10 \times 10 \times 12.9\,\text{mm}^3$ are excised with a bandsaw. For tension testing, dogbone specimens are cut using a 75W CO_2 laser cutter (Universal Laser Systems Inc., Scottsdale, AZ) to the ASTM D3574 standard geometry for tensile testing of foam polymers. For indentation experiments (Section 4.1), $25.4 \times 25.4 \times 12.9$ mm^3 samples are laser cut and bonded to an aluminum backer on one of the large faces using a two-part epoxy (ITW Polymers Adhesives, Danvers, MA). For each simple-shear-like experiment (Section 4.2), two $30 \times 30 \times 12.9\,\text{mm}^3$ specimens are laser cut, and each large, square face is epoxied to an aluminum backer. Finally, circular holes are cut out of tension specimens to be used for the third type of validation testing (Section 4.3). DIC measurements

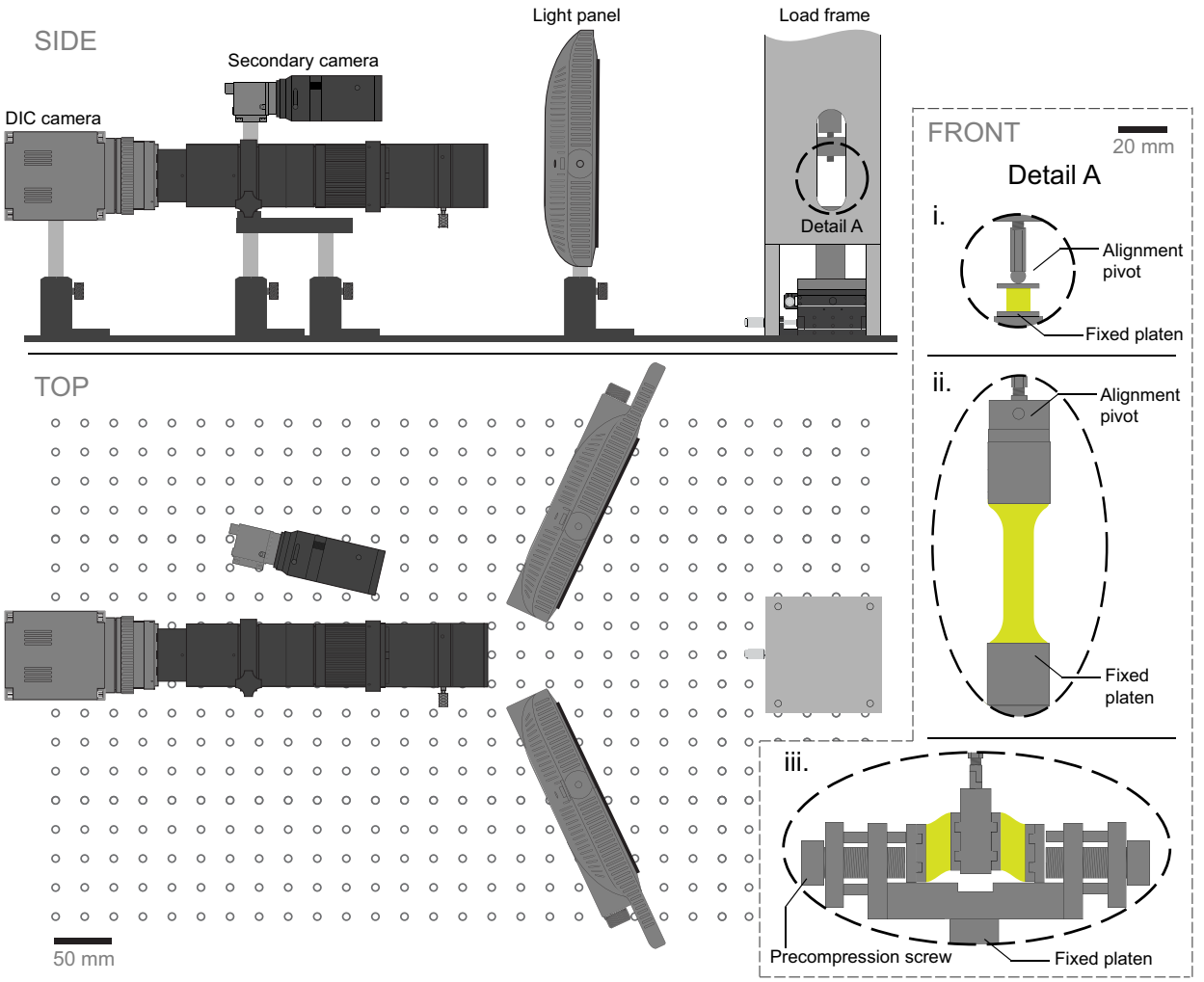


Fig. 3. Schematic of the top and side views of the experimental setup as deployed on a vibration-isolated optical breadboard. The arrangement of the two cameras with lenses, light panels with linearly polarizing filters, and load frame is shown. Detail A gives front-view diagrams (i.e., as seen by the cameras) of the fixtures used for (i) simple compression, (ii) simple tension, and (iii) simple-shear-like deformation.

are used to obtain kinematic data in all experiments except for indentation testing, and to prepare specimens for DIC analysis, a face orthogonal to the testing axis is speckle-painted for each specimen. To speckle the specimens, a black base coat of acrylic ink (Liquitex Artist Materials, Piscataway, NJ) is airbrushed (HP-C Plus, Iwata Medea Inc, Portland, OR) on the surface, followed by a white speckle layer (Golden Artist Colors Inc., New Berlin, NY). For each configuration, the optical train employed is configured such that this technique yields speckles of approximately 3–7 pixels in size. Due to the inherent DIC challenges posed by mechanical testing of elastomeric foams – namely, that large speckle distortions and speckle loss due to cell collapse or opening arise when the material undergoes finite deformations – a custom, DIC procedure is utilized (Landauer et al., 2018), which incorporates the concept of cross-correlation quality factors to improve the accuracy of DIC measurements involving distorted speckle patterns undergoing large deformations. Finally, for all experiments, both characterization and validation, three specimens are tested, and each specimen is tested three times (i.e., nine total experiments per condition). For repeated experiments on the same specimen, we ensure that the wait-time between experiments is sufficiently long to allow complete relaxation of any time-dependent material physics, typically 3 h or more in total test time.

Experimental setup. A schematic layout of the optical system and loading apparatus used in our experiments is shown in Fig. 3. For all experiments, the loading apparatus and control software are the same as that described in Landauer et al. (2018) – namely, a screw-driven linear actuator (Ultra Motion, Cutchogue, NY) and a load cell (LCFD-50, Omega Engineering, Stamford, CT) with a custom LabView (National Instruments, Austin, TX) interface that allows synchronized actuator displacement control, actuator displacement measurement, imaging, and force data collection. The optical system utilized to collect data for two-dimensional DIC is comprised of a custom control computer, a scientific camera

(edge5.5, PCO-Tech Inc., Romulus, MI), a long-distance microscopy lens (K2/S, Infinity Photo-Optical Company, Boulder, CO), polarizing filters (American Polarizers Inc., Reading, PA and Edmund Optics, Barrington, NJ), and LED light panels (Lykos Daylight, Manfrotto, South Upper Saddle River, NJ), which is assembled on the same vibration-isolated table as the loading apparatus. Additionally, for all experiments, a secondary imaging camera with a field-of-view encompassing the complete specimen and load frame is used to detect any potential anomalous loading conditions (e.g., onset of specimen failure at the grips).

For each experiment type, a custom load-frame insert is used to grip specimens during loading, and the three primary fixtures – simple compression, simple tension, and simple-shear-like deformation – are diagrammed in the insets of Fig. 3. By using modular inserts, the control procedures and DIC parameters remain consistent across experiment types. For simple compression, the insert consists of graphitic steel top and bottom platens compressed via a spherical indenter tip attached to the load cell. The spherical indenter tip acts as a pivot point, allowing any misalignment to be identified and eliminated in our compression experiments. The contact areas between the platens and the specimen are lubricated with graphite power and spherical glass microbeads (diameter 20–30 μm). For simple tension, screw-actuated wedge-and-pin-type grips – with the gripping-force screw acting as the pin – are attached to the base of the load frame and to the load cell. For the simple-shear-like validation experiments, a symmetric "double-shear" insert, involving two specimens with aluminum backers epoxied to both sides, is utilized, as shown in inset (iii) of Fig. 3. The center platen undergoes a vertical displacement prescribed by the linear actuator to shear the specimens. Additionally, horizontal screws on each side of the fixture may be adjusted to subject the specimens to a uniaxial pre-compression, which we set to be equal for both specimens in order to maintain symmetry, and which is held constant during the subsequent shear deformation.

Details of simple compression/tension tests. The characterization experiments, i.e., simple compression and tension, are conducted in displacement control to engineering strain limits of -0.75 (axial stretch of $\lambda = 0.25$) in compression and 0.5 (axial stretch of $\lambda = 1.5$) in tension and are loaded and unloaded at a constant, quasi-static engineering strain-rate magnitude of 2.0×10^{-4} s^{-1}.[2] Illustrative example images of reference and deformed specimens for the high density foam are shown in Fig. 4(a) and (b) for compression and tension, respectively, where the direction of compression/tension is identified as the x_1-direction, and the selected regions-of-interest for DIC analysis are outlined with dashed lines. Contours of the resultant two-dimensional displacement fields – i.e., $u_1(x_1, x_2)$ and $u_2(x_1, x_2)$ – at sample levels of deformation, namely an axial stretch of $\lambda = 0.35$ for compression and $\lambda = 1.50$ for tension, calculated using the custom qDIC procedure are also shown in Fig. 4. In both examples, the contours reveal an axial displacement field u_1 that depends linearly on the axial referential coordinate x_1 and a lateral displacement field u_2 that depends linearly on the lateral referential coordinate x_2 – indicating that the intended homogeneous deformation without shearing and with minimal barreling has been achieved. Importantly, in our compression experiments, we do not observe evidence of strain localization into compaction bands, such as those that often arise during compression of very-low-density, "reticulated" foams (e.g., Elliott et al., 2002; Lakes et al., 1993; Wang et al., 2000). This is quantified via the histogram-based technique of Wang and Cuitiño (2002) (see Appendix C). Furthermore, at the spatial resolution of the DIC algorithm using a subset size of 16×16 px^2 (see Appendix C), inhomogeneities in the displacement field are an order of magnitude smaller than displacement due to the globally applied deformation and of similar scale as the displacement noise floor.

Components of the global displacement gradient tensor are extracted from the displacement fields for each deformation step in each experiment via a global plane-fitting procedure, which is described in detail in Landauer et al. (2018), and the components du_1/dx_1 and du_2/dx_2 are identified as the axial engineering strain and lateral engineering strain, respectively. The corresponding axial engineering stress is computed from load-cell data and the previously-measured, referential cross-sectional area. A summary of the experimental data-flow used to reduce experimental data to axial engineering strain, lateral engineering strain, and axial engineering stress data is outlined in Fig. 5. The axial engineering stress versus axial engineering strain and lateral engineering strain versus axial engineering strain data are plotted as solid lines in Fig. 1 for the (a) low, (b) moderate, and (c) high density foams. The black curves represent the average of the nine instances of each experiment, and the shaded areas surrounding the solid curves represent one standard deviation above and below the average, which indicate good repeatability in our experiments. Arrows indicate loading and unloading, demonstrating that hysteresis is minimal in our experiments and that the strain-rate is sufficiently low so that the experimentally-measured behavior represents the equilibrium, elastic response of the material. Finally, we note that characterization of the foam material in simple compression along each of the three orthogonal directions of the as-received sheets revealed no discernible difference in the experimentally-measured behavior. This observation, along with the random microstructure of the foam material (Fig. 2), supports our choice to model the foam material as isotropic.

3. Constitutive theory

In this section, we describe our hyperelastic constitutive model for the elastic behavior of isotropic, elastomeric foams and apply the model to three densities of Poron XRD foam.

[2] The choice of quasi-static strain-rate was informed by preliminary stress relaxation experiments on the moderate density foam.

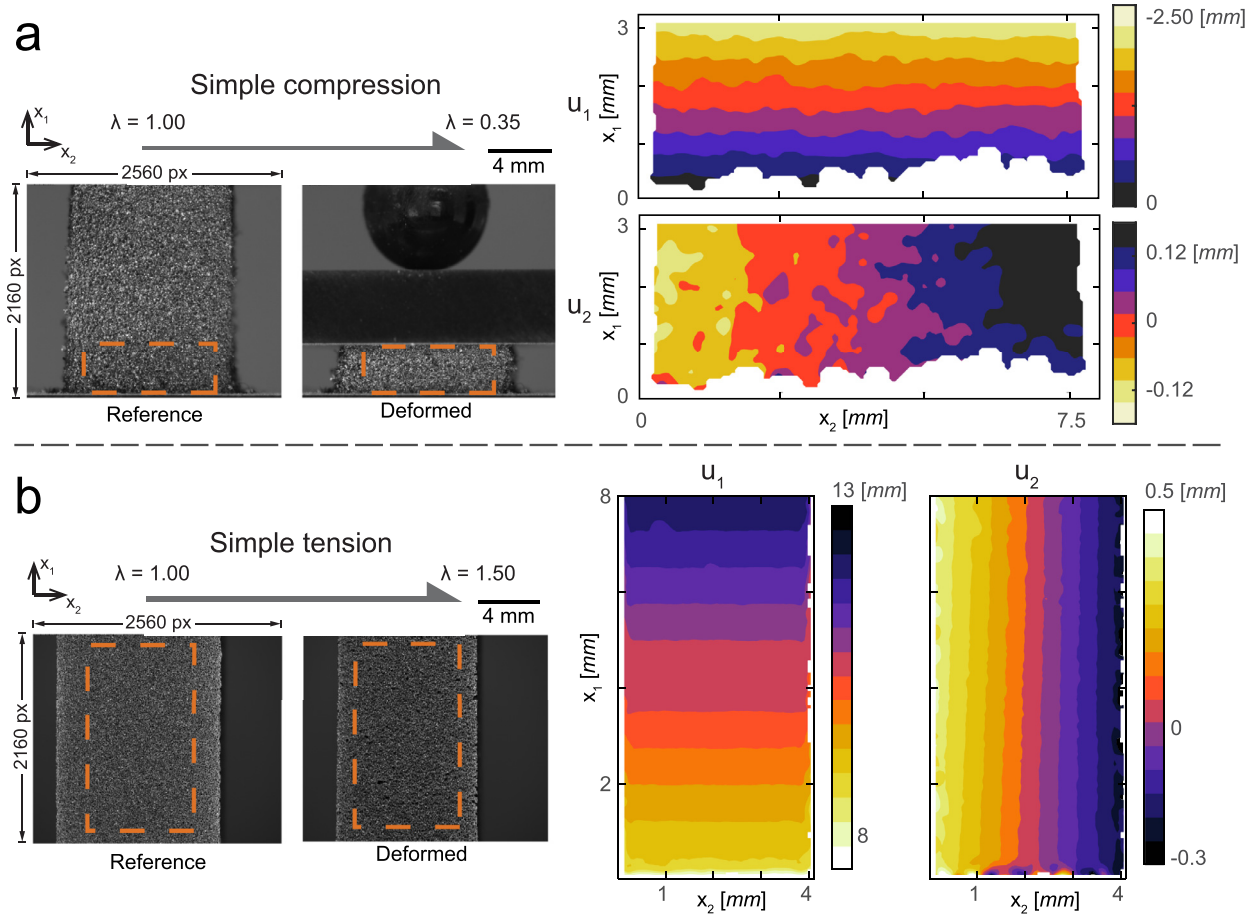


Fig. 4. Example kinematic data collected from (a) simple compression (adapted from Landauer et al., 2018) and (b) simple tension experiments on the high density foam. (Left) Images of the respective specimens in the reference and deformed states with the regions-of-interest for DIC analysis outlined with dashed lines. DIC regions are approximately 1200×350 px^2 for compression and approximately 800×1900 px^2 for tension. (Right) Contours of the DIC-calculated, two-dimensional displacement fields for the axial (u_1) and lateral (u_2) displacement components.

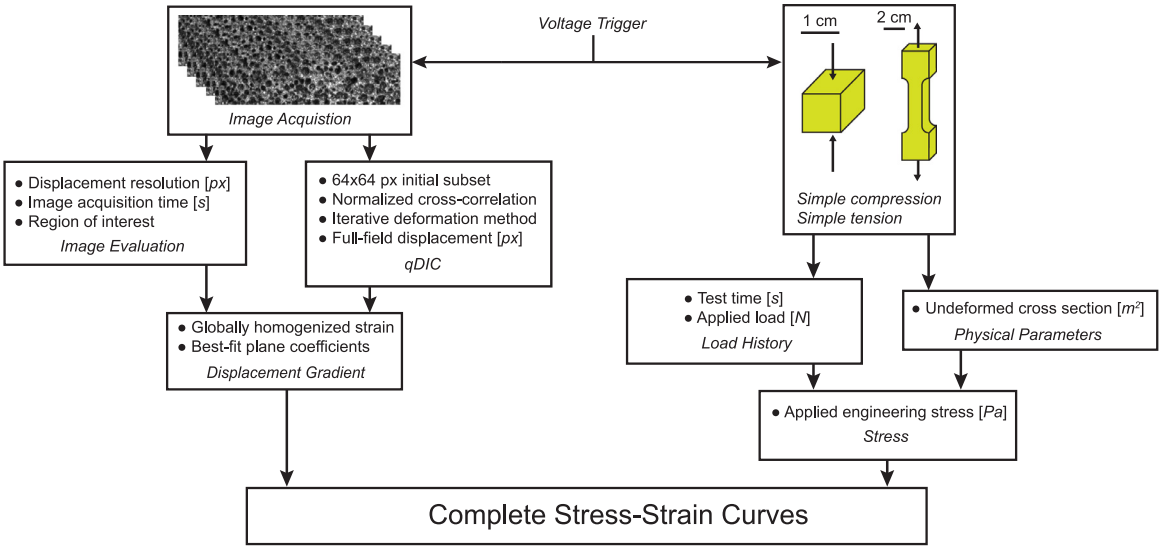


Fig. 5. Outline of the experimental data-flow for quality-factor-based digital image correlation (qDIC) data, which is reduced to axial and lateral engineering strain data, and force data, which is used to compute the axial engineering stress. Experiments are initiated via a voltage signal provided when the imposed displacement history is begun, ensuring that time-synchronized load and images datasets are collected.

Basic kinematics. We consider a body B identified with the region of space it occupies in a fixed reference configuration and denote an arbitrary material point within B as $\mathbf{x}$. The referential body B then undergoes a motion $\mathbf{y} = \boldsymbol{\chi}(\mathbf{x}, t)$ to the deformed body $\mathcal{B}_t$ at each time t. The deformation gradient is given by $\mathbf{F} = \nabla\boldsymbol{\chi}$, such that the ratio of referential to deformed volume is strictly greater than zero, i.e., $J = \det\mathbf{F} > 0$.[3] The displacement field is denoted as $\mathbf{u}(\mathbf{x}, t) = \boldsymbol{\chi}(\mathbf{x}, t) - \mathbf{x}$ and the left and right Cauchy-Green deformation tensors are $\mathbf{B} = \mathbf{F}\mathbf{F}^\top$ and $\mathbf{C} = \mathbf{F}^\top\mathbf{F}$, respectively. The deformation tensors admit the spectral decompositions $\mathbf{B} = \sum_{i=1}^{3}\lambda_i^2\mathbf{l}_i \otimes \mathbf{l}_i$ and $\mathbf{C} = \sum_{i=1}^{3}\lambda_i^2\mathbf{r}_i \otimes \mathbf{r}_i$, where $\{\lambda_i | i = 1, 2, 3\}$ are the principal stretches, $\{\mathbf{l}_i | i = 1, 2, 3\}$ are the principal directions of $\mathbf{B}$, and $\{\mathbf{r}_i | i = 1, 2, 3\}$ are the principal directions of $\mathbf{C}$. Finally, we introduce the logarithmic (Hencky) finite-strain tensor in the deformed body: $\mathbf{E} = \sum_{i=1}^{3}(\ln\lambda_i)\mathbf{l}_i \otimes \mathbf{l}_i$.

An isotropic free-energy function. The constitutive response of a hyperelastic material is specified through the free-energy density function ψ. In general, ψ is a function of the right Cauchy-Green tensor $\mathbf{C}$, i.e., $\psi = \hat{\psi}(\mathbf{C})$, but for an isotropic material, the free-energy function is an isotropic function of $\mathbf{C}$ and may be represented as a function of the principal stretches or three independent invariants. Stretch-based, Ogden-type free-energy functions of the form $\psi = \tilde{\psi}(\lambda_1, \lambda_2, \lambda_3)$ are commonly used to model both incompressible (Ogden, 1972b) and compressible (Ogden, 1972a; Storakers, 1986) elastomers; however, as mentioned in Section 1, this approach can be unwieldy to work with for highly-compressible materials. Alternatively, invariant-based free-energy functions utilizing the principal invariants of $\mathbf{C}$, i.e., $I_1 = \mathrm{tr}\,\mathbf{C}$, $I_2 = (1/2)((\mathrm{tr}\,\mathbf{C})^2 - \mathrm{tr}\,(\mathbf{C}^2))$, $I_3 = \det\mathbf{C} = J^2$, may be invoked, and indeed, several hyperelastic models for porous elastomers fall in this category (e.g., Anani and Alizadeh, 2011; Blatz and Ko, 1962; Danielsson et al., 2004; Shrimali et al., 2019; Yang and Shim, 2004). However, developing free-energy functions of the form $\psi = \tilde{\psi}(I_1, I_2, I_3)$ that depend on all three invariants, which is necessary to capture experimental data for compressible foams, is difficult, since it is not straightforward to isolate the effect of each invariant.[4] To address this challenge, we develop a free-energy function that depends on invariants of the logarithmic strain tensor $\mathbf{E}$, as proposed by Criscione et al. (2000). The logarithmic-strain invariants of Criscione et al. (2000), which we denote as $\{K_1, K_2, K_3\}$, represent specific aspects of deformation, enabling a more tractable path to introducing phenomenological fitting functions that capture experimental data. The three invariants are defined as follows:

1. The first invariant is $K_1 = \mathrm{tr}\,(\mathbf{E}) = \ln(J) = \ln(\lambda_1\lambda_2\lambda_3)$ and represents the volume change of the material with positive values for dilatation and negative values for compaction.
2. The second invariant is $K_2 = |\mathrm{dev}\,(\mathbf{E})| \geq 0$ and represents the magnitude of constant-volume distortion in the material. In terms of the principal stretches, the second invariant is given by $K_2 = \sqrt{(\ln\lambda_1 - (1/3)K_1)^2 + (\ln\lambda_2 - (1/3)K_1)^2 + (\ln\lambda_3 - (1/3)K_1)^2}$. Defining the tensorial direction of deviatoric strain as $\mathbf{N} = \mathrm{dev}\,(\mathbf{E})/K_2$, such that $\mathbf{N}$ is a deviatoric and unit-magnitude tensor, the logarithmic strain may be expressed as $\mathbf{E} = \frac{1}{3}K_1\mathbf{1} + K_2\mathbf{N}$.
3. The third invariant is $K_3 = 3\sqrt{6}\det(\mathbf{N})$ and represents the mode of distortion. The pre-factor $3\sqrt{6}$ is present so that $-1 \leq K_3 \leq 1$. The third invariant is constant for a given distortion mode (e.g., $K_3 = -1$ in simple compression, $K_3 = 0$ in simple shear, and $K_3 = 1$ in simple tension) and hence, K_3 remains fixed during the simple compression/tension experiments utilized for material characterization. In terms of the principal stretches, the third invariant may be expressed as $K_3 = 3\sqrt{6}(\ln\lambda_1 - (1/3)K_1)(\ln\lambda_2 - (1/3)K_1)(\ln\lambda_3 - (1/3)K_1)/K_2^3$.

As shown by Criscione et al. (2000), the Cauchy stress is then given through the derivatives of the free-energy function $\psi = \tilde{\psi}(K_1, K_2, K_3)$ by

$$\mathbf{T} = J^{-1}\left[\frac{\partial\tilde{\psi}}{\partial K_1}\mathbf{1} + \frac{\partial\tilde{\psi}}{\partial K_2}\mathbf{N} + \frac{\partial\tilde{\psi}}{\partial K_3}\frac{1}{K_2}\mathbf{Y}\right] \tag{3.1}$$

where $\mathbf{Y} = 3\sqrt{6}\mathbf{N}^2 - \sqrt{6}\mathbf{1} - 3K_3\mathbf{N}$.[5]

Decomposition of the free-energy function. We take the free-energy density function $\tilde{\psi}(K_1, K_2, K_3)$ to be additively decomposed as follows:

$$\tilde{\psi}(K_1, K_2, K_3) = G_0\left[X(K_1)K_2^2 + L(K_2, K_3)\right] + Bf(K_1), \tag{3.2}$$

where G_0 and B are the ground-state shear and bulk moduli, respectively, and $X(K_1)$, $L(K_2, K_3)$, and $f(K_1)$ are phenomenological fitting functions. The interpretation of each term is as follows:

1. For a material undergoing large volumetric deformation, interaction between the volumetric and distortional contributions to the free energy is expected. The $G_0X(K_1)K_2^2$ term characterizes this coupling between the volumetric and distortional responses. For simplicity, we assume that this term only involves low-order K_2-dependence, i.e., that it only

[3] *Notation:* The symbols ∇, Div and Curl denote the gradient, divergence, and curl with respect to the material point $\mathbf{x}$ in the reference body B; grad, div, and curl denote these operators with respect to the point $\mathbf{y} = \boldsymbol{\chi}(\mathbf{x}, t)$ in the deformed body $\mathcal{B}_t$. We write $\mathrm{tr}\,\mathbf{A}$, $\det\mathbf{A}$, and $\mathrm{dev}\,\mathbf{A}$ to denote the trace, determinant, and deviatoric part of a tensor $\mathbf{A}$, respectively.

[4] See, for example, Criscione (2004) for a discussion of this issue in the context of invariant-based free-energy functions for incompressible materials.

[5] Eq. (3.1) may be obtained by recognizing that for an isotropic, hyperelastic material, the Kirchhoff stress $\mathbf{T}_\mathrm{K} = J\mathbf{T}$ is given through the derivative of the free-energy function with respect to the spatial logarithmic strain tensor – i.e., $\mathbf{T}_\mathrm{K} = \partial\tilde{\psi}/\partial\mathbf{E}$ – and then utilizing the chain rule along with the identities $\partial K_1/\partial\mathbf{E} = \mathbf{1}$, $\partial K_2/\partial\mathbf{E} = \mathbf{N}$, and $\partial K_3/\partial\mathbf{E} = \mathbf{Y}/K_2$.

depends on K_2 through K_2^2, and is independent of the third invariant K_3. The volumetric dependence of this term may be more general through the function $X(K_1)$. More complex forms of volumetric/distortional coupling may certainly be invoked, but we find that this simple form is capable of capturing experimental data while enabling a streamlined material parameter estimation procedure – discussed further in Appendix A. The function $X(K_1)$ may be interpreted as encompassing the volumetric-strain-dependence of the instantaneous shear modulus. We require that $X(K_1 = 0) = 1$, and for the purpose of stability – discussed further in Section 5 – we also require that the function $X(K_1)$ remain positive over the intended K_1-range of application.

2. The $G_0 L(K_2, K_3)$ term represents the higher-order distortional response, i.e., higher-order than the K_2^2-dependence of the first term, which we take to be uncoupled from the volumetric response for simplicity. This term is intended to capture stiffening behavior as the magnitude of distortion, K_2, increases, and the inclusion of K_3-dependence allows for mode-dependence of the higher-order distortional response. We require that $L(K_2 = 0, K_3) = \partial L/\partial K_2|_{K_2=0} = 0$, and that, following Criscione et al. (2000), $\partial L/\partial K_3$ goes to zero as order K_2^3 or higher as K_2 goes to zero. Moreover, we require that $\partial L/\partial K_2 > 0$ for all $K_2 > 0$ and $-1 \le K_3 \le 1$, so that the function $L(K_2, K_3)$ increases monotonically with K_2.
3. Finally, $Bf(K_1)$ describes the purely volumetric response. We require that $f(K_1 = 0) = 0$ and $df/dK_1|_{K_1=0} = 0$. Furthermore, for the purpose of stability, we require that $d^2 f/dK_1^2 > 0$ for all K_1, i.e., that df/dK_1 is a monotonically increasing function of K_1.

Using (3.2) in (3.1), we obtain the following expression for the Cauchy stress:

$$\mathbf{T} = J^{-1}\left[\left(G_0 \frac{dX}{dK_1} K_2^2 + B\frac{df}{dK_1}\right)\mathbf{1} + G_0\left(2X(K_1)K_2 + \frac{\partial L}{\partial K_2}\right)\mathbf{N} + G_0 \frac{\partial L}{\partial K_3}\frac{1}{K_2}\mathbf{Y}\right]. \tag{3.3}$$

Note that the first term in (3.3), namely, the term involving $G_0(dX/dK_1)K_2^2\mathbf{1}$, represents a shear-induced mean stress that arises due to the coupled volumetric/distortional response.

Specialization of the free-energy function. To fit to the experimental data collected in Section 2 for low, moderate, and high density Poron XRD, we choose specific forms for the phenomenological functions appearing in (3.2). We emphasize that, while these forms are intended to capture data for Poron XRD, future modification may be necessary to apply the modeling approach to other foam materials.

1. Based on experimental data, we find that a form for the function $X(K_1)$ that linearly depends on K_1 but with different slopes before and after the onset of the plateau regime in compression is sufficient to capture the response. Accordingly, we adopt the following simple phenomenological form for dX/dK_1:

$$\frac{dX}{dK_1} = \frac{1}{2}\left(X_1' + X_2'\right) + \frac{1}{2}\left(X_1' - X_2'\right)\tanh\left[\frac{1}{\Delta_K}(K_1 - K_1^0)\right], \tag{3.4}$$

where $K_1^0 < 0$ is the value of K_1 denoting the transition to the plateau regime, $\Delta_K > 0$ is a parameter denoting the K_1-range across which the assumed hyperbolic-tangent-type transition in the value of dX/dK_1 transitions from X_1' prior to the plateau regime in compression (i.e., $K_1 \gtrsim K_1^0$) to X_2' after the onset of the plateau regime in compression (i.e., $K_1 \lesssim K_1^0$). To capture experimental data, the value of X_2' is small compared to X_1'. Integrating (3.4) with respect to K_1 and applying the condition $X(K_1 = 0) = 1$, we obtain the following phenomenological form for $X(K_1)$:

$$X(K_1) = \frac{1}{2}\left(X_1' + X_2'\right)K_1 + \frac{\Delta_K}{2}\left(X_1' - X_2'\right)\ln\left[\frac{\cosh\left((K_1 - K_1^0)/\Delta_K\right)}{\cosh\left(K_1^0/\Delta_K\right)}\right] + 1. \tag{3.5}$$

2. The higher-order distortional response is taken to be given through

$$L(K_2, K_3) = C_0 K_2^p + C_1(1 + K_3)K_2^q, \tag{3.6}$$

where $C_0 > 0$ and $C_1 > 0$ are constant parameters and $p > 2$ and $q > 2$ are higher-order exponents. We note that since $K_3 = -1$ in compression, the second term does not affect the response in compression. This term is present to allow for different higher-order responses in compression and tension to be specified.
3. Finally, the purely volumetric response function is fitted to

$$\frac{df}{dK_1} = \frac{J^{C_2} - 1}{C_2} + C_3 J\left[J^{-r} - \left(\frac{1 - J_{\min}}{J - J_{\min}}\right)^r\right], \tag{3.7}$$

where $J = \exp(K_1)$ is the volume ratio and $J_{\min}$, r, C_2, and C_3 are constant parameters. We note that the parameter $J_{\min}$ represents the minimum allowable value of J and that df/dK_1 diverges as J approaches $J_{\min}$ from above. The parameter $J_{\min}$ is limited to be within the range $0 < J_{\min} < 1$ and may be interpreted as $J_{\min} = 1 - \phi_0$, where ϕ_0 is the void volume fraction of the undeformed foam. The first term in (3.7) is intended to capture the response in tension and prior to the plateau regime in compression, while the second term is intended to capture the rapidly stiffening behavior in compression. We note that $d^2 f/dK_1^2|_{K_1=0}$ does not precisely equal one. However, since the first term is intended to capture the response prior to the onset of the plateau regime, its contribution will dominate the second derivative at $K_1 = 0$, so that

Table 1
Material parameters for low, moderate, and high density Poron XRD foams. In total, fourteen parameters are utilized, with four density-dependent parameters and ten density-independent parameters.

Density	Density-dependent				Density-independent									
	G_0 [kPa]	B [kPa]	$J_{\min}$	C_1	K_1^0	Δ_K	X_1'	X_2'	C_0	p	q	r	C_2	C_3
Low	34.5	58.7	0.12	2.5										
Moderate	65.2	117.4	0.16	1.9	-0.21	0.2	3.7	0.22	0.1	4	5	2	9	0.026
High	102.0	193.8	0.19	1.9										

$d^2 f/dK_1^2|_{K_1=0} \approx 1$, and B approximately represents the ground-state bulk modulus.[6] Upon integrating (3.7) with respect to K_1 (for $r \neq 1$) and applying the condition $f(K_1 = 0) = 0$, we obtain the following functional form for f:

$$f = \frac{J^{C_2} - C_2 \ln J - 1}{C_2^2} + \frac{C_3}{r-1}\left[\frac{-1}{J^{r-1}} + \frac{(1-J_{\min})^r}{(J-J_{\min})^{r-1}} + J_{\min}\right]. \tag{3.8}$$

Material parameters for Poron XRD. We estimate the material parameters appearing in our constitutive model – a total of fourteen parameters – for Poron XRD from the simple compression/tension data in Fig. 1. For ease of presentation, the discussion of our heuristic procedure for material parameter estimation is relegated to Appendix A. The material parameters for all three densities of Poron XRD determined using this procedure are listed in Table 1. Four of the material parameters are density-dependent – G_0, B, $J_{\min}$, and C_1 – and the remaining ten parameters are density-independent and are not refit for each of the three foam densities. Fig. 1 compares simple compression/tension experimental data (solid lines) with corresponding calculations using the fitted model (dashed lines). The quality of the comparison between the experimental data and the fitted model for both the axial engineering stress and the lateral engineering strain data and for all three densities is quite reasonable.

4. Validation experiments and simulations

In this section, we validate the predictive capability of our hyperelastic constitutive theory by comparing model predictions against experimental measurements in settings involving inhomogeneous deformation. To probe the model in a variety of situations, i.e., compression, shear, and tension-dominated situations, we consider three specific types of validation tests: (1) spherical and conical indentation, (2) simple-shear-like deformation both without and with a fixed amount of pre-compression, and (3) tension of a specimen with circular holes. The constitutive model of Section 3 has been numerically implemented in Abaqus/Standard (Abaqus, 2018) using a user-material (UMAT) subroutine, and throughout this section, we obtain model predictions in inhomogeneous deformation scenarios using finite-element calculations in Abaqus/Standard.

4.1. Spherical and conical indentation

We begin by considering indentation – a validation case that involves compression-dominated deformation directly beneath the indenter – for both spherical and conical indenter geometries. As discussed in Section 2, indentation experiments are conducted on foam specimens with a square $25.4 \times 25.4\,\text{mm}^2$ cross section and a height of $H = 12.9$ mm. One square face of each specimen is constrained by a bonded aluminum backer, and the specimen is indented on the opposite unconstrained face. Spherical indentation is performed using a steel indenter tip with a radius of $r = 3.97$ mm, and conical indentation is performed using an aluminum indenter tip with a base angle of $\theta = 30^\circ$. Photographs of the experimental setup prior to deformation for spherical and conical indentation are shown in Fig. 6(a) and (c), respectively. Experiments are performed at an indenter displacement-rate of $\dot{\delta} = 2.0 \times 10^{-3}$ mm/s for spherical indentation and $\dot{\delta} = 2.5 \times 10^{-3}$ mm/s for conical indentation. The maximum indenter displacements achieved during spherical and conical indentation are around 4 mm in all cases. The experimentally-measured indenter force P versus displacement δ relations for spherical and conical indentation are shown as solid lines in Fig. 6(b) and (d), respectively. In both cases, data for all three foam densities are shown. As in Fig. 1, the shaded regions surrounding the solid curves represent one standard deviation above and below the average across all nine instances of the experiment, and the arrows indicate loading and unloading. Hysteresis is minimal, demonstrating that the chosen indenter displacement-rates are sufficiently slow.

For the corresponding finite-element simulations, we idealize the geometry as axisymmetric for computational efficiency. This involves idealizing the cuboid foam specimen as a disk of radius R_0 and height $H = 12.9$ mm. The outer radius is set to be $R_0 = 14.4$ mm, which is the average value of the outer radius of the square specimen over the angular coordinate.[7]

[6] Using (3.7), the exact expression for the ground-state bulk modulus is $B_0 = B(d^2 f/dK_1^2|_{K_1=0}) = B[1 + C_3 r J_{\min}/(1 - J_{\min})]$. To fit the rapidly stiffening behavior in compression, we take $C_3 \ll 1$, and hence, $B_0 \approx B$.

[7] We have performed selected three-dimensional simulations, in which the experimental geometry is exactly represented, and the corresponding simulated indenter force versus displacement result is indistinguishable from the axisymmetric result, verifying that the axisymmetric idealization is valid.

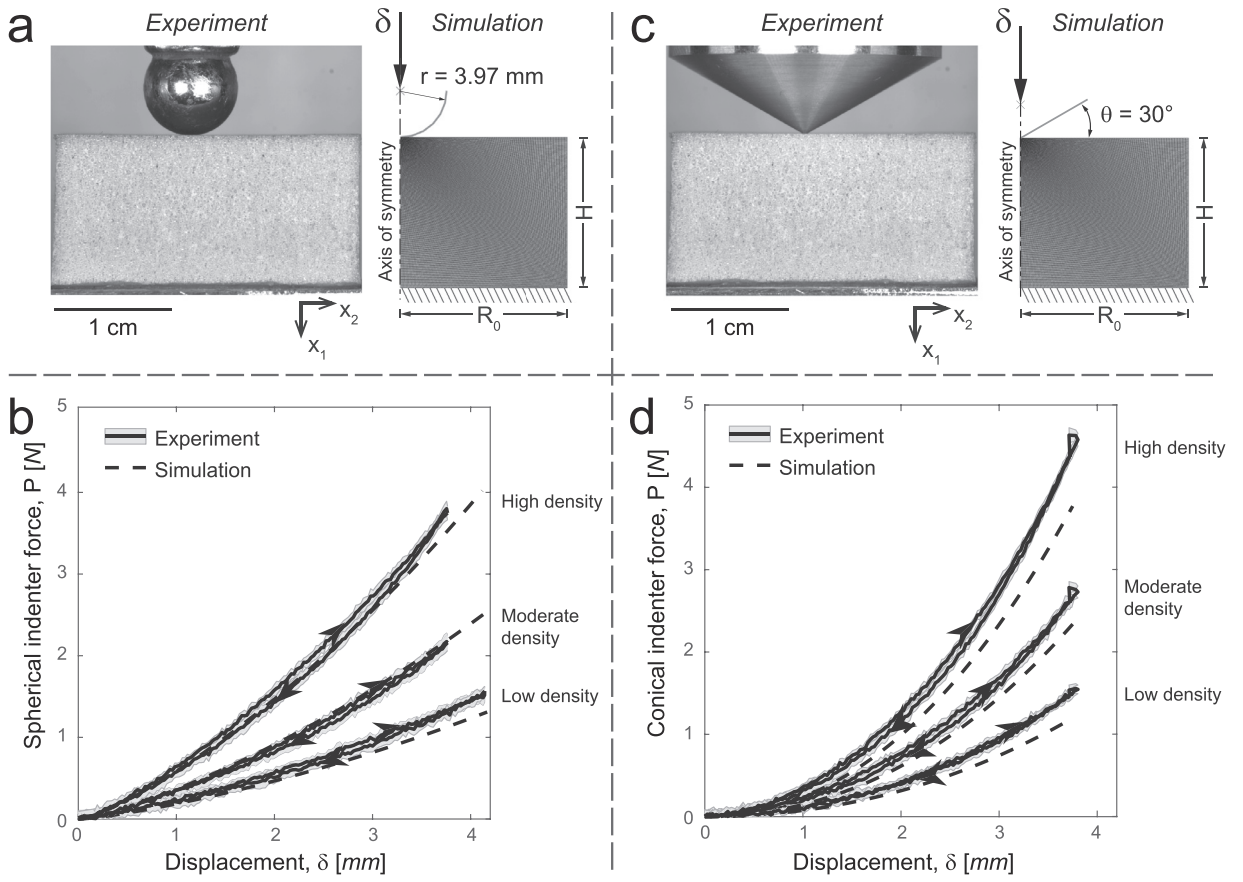


Fig. 6. Spherical and conical indentation validation tests. Snapshots of the experimental setup prior to deformation and the corresponding finite-element configurations are given for (a) spherical and (c) conical indentation. Indenter force P versus indenter displacement δ for (b) spherical and (d) conical indentation. Solid lines with shaded error regions represent experimental data, and dashed lines are the predictions from the constitutive model.

The axisymmetric finite-element configurations for spherical and conical indentation are shown in Fig. 6(a) and (c), respectively. The foam substrate is meshed using 17,236 Abaqus-CAX4H elements (four-node, axisymmetric, quadrilateral, constant-pressure elements) and the indenters are idealized as rigid surfaces. Contact between the foam substrate and the indenter is approximated as rough – i.e., no slip – in both cases.[8] Regarding boundary conditions, to model the fixed-base boundary condition, all displacement components are prescribed to be zero on the bottom face of the mesh. The left face of the mesh is coincident with the axis of symmetry, and hence, the horizontal displacement is set to zero along this face. The right face of the mesh is left unconstrained. The calculated indenter force P versus displacement δ relations – obtained without any parameter adjustment – are included as dashed lines for all three foam densities in Fig. 6(b) and (d). Generally, there is good agreement between the experimentally-measured and simulated responses, especially for the case of spherical indentation.[9]

4.2. Simple-shear-like deformation without and with pre-compression

Next, we validate aspects of the model related to shear-dominated deformation. We perform two types of experiments, carried out on foam cuboids with a square cross-section of side-length $W = 30$ mm and a height of $H = 12.9$ mm. First, we subject foam cuboids to simple-shear-like deformation, in which one square face is translated parallel to the other by a shear displacement δ. In these experiments, the square faces are constrained by bonded aluminum backers, and the shear displacement is applied by moving one backer at a rate of $\dot{\delta}/H = 2.0 \times 10^{-4}$ s^{-1} to a maximum shear displacement of $\delta/H = 0.5$ and then unloading, while the other backer remains fixed, and the distance between the backers is held at H.

[8] We have performed corresponding simulations with frictionless interaction between the foam substrate and the indenter, and the difference between the rough and frictionless cases is negligible, indicating that the role of surface friction is small in indentation of compressible foam materials.

[9] We note that for the conical indentation case, the indenter forces predicted by the model are slightly lower than the corresponding experimental data, which may be rationalized as follows. To optimize the stability of the model, the fitted volumetric response function, df/dK_1, slightly underestimates experimental data. The under-prediction in the conical indenter force is associated with the large volumetric deformations that occur during conical indentation, as compared to spherical indentation, which is slightly less volumetrically-dominated.

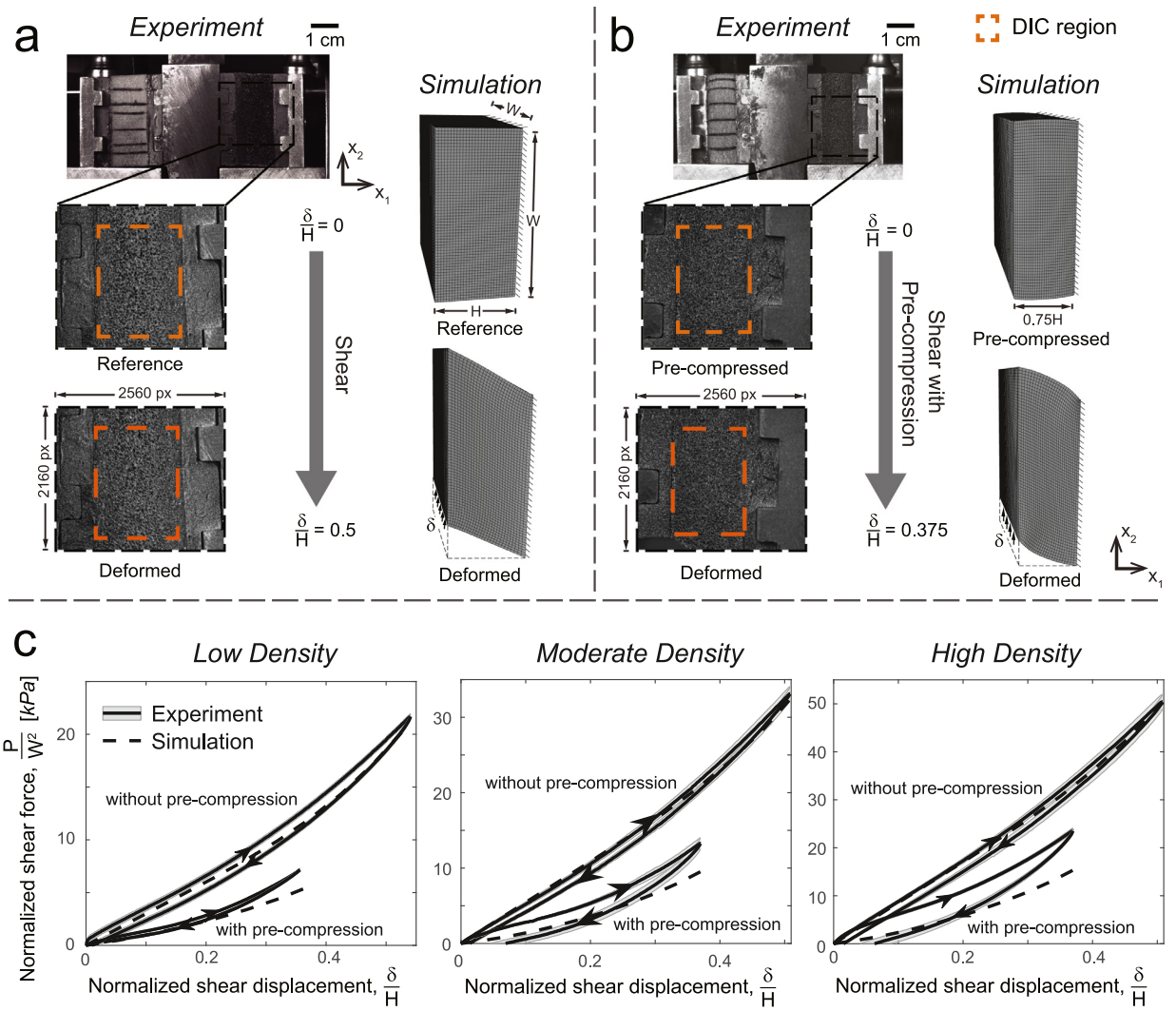


Fig. 7. Simple-shear-like validation tests both without and with uniaxial pre-compression. (a) Snapshot of the double-shear fixture with two specimens prior to deformation and the experimental region-of-interest used for DIC imaging (approximately 1300×1200 px^2) in both the reference and the deformed states for the case without pre-compression. The corresponding finite-element configuration is shown on the right, also in both the reference and the deformed states. (b) Experimental images and finite-element configuration for the case with pre-compression (DIC region approximately 950×1200 px^2), displaying the pre-compressed and the deformed, sheared states. (c) Normalized shear force P/W^2 versus normalized shear displacement δ/H both without and with pre-compression and for all three densities. Solid lines with shaded error regions represent experimental data, and dashed lines are simulated predictions of the constitutive model.

Notably, all rectangular side-faces remain unconstrained, so that edge-effects are present, and hence, deformation is simple-shear-like but inhomogeneous. In the second type of experiment, we consider simple-shear-like deformation with a fixed uniaxial pre-compression as an example of coupled distortional/volumetric deformation. The simple-shear-like experiment with pre-compression is carried out in two steps. First, the specimen is compressed, such that the deformed distance between the constrained faces is 0.75H. Specimens are then allowed to rest for at least 40 min, while the pre-compression is held fixed in order to allow complete relaxation of viscoelastic material behavior. Second, the specimen is sheared by a shear displacement δ at the same rate of $\dot{\delta}/H = 2.0 \times 10^{-4}$ s^{-1} to a maximum shear displacement of $\delta/H = 0.375$ and then unloaded, while the distance between the constrained faces is held fixed at 0.75H. In both types of experiments, we utilize the double-shear fixture pictured in Fig. 7(a), which is akin to a double-lap or three-bar (e.g., ASTM D4255 Procedure B) shear design. In this configuration, two specimens are tested simultaneously in order to maintain symmetry, and therefore, the resultant shear force P is identified as half the force applied to the moving center platen. The measured normalized shear force P/W^2 versus shear displacement δ relations for all three foams densities and both without and with pre-compression are shown as solid lines in Fig. 7(c). As before, the shaded regions denote the range of one standard deviation above and below the average experimental measurement, and the arrows denote loading and unloading, demonstrating minimal hysteresis. Two features are notable in the force-displacement relations. First, in shear-dominated deformation, no

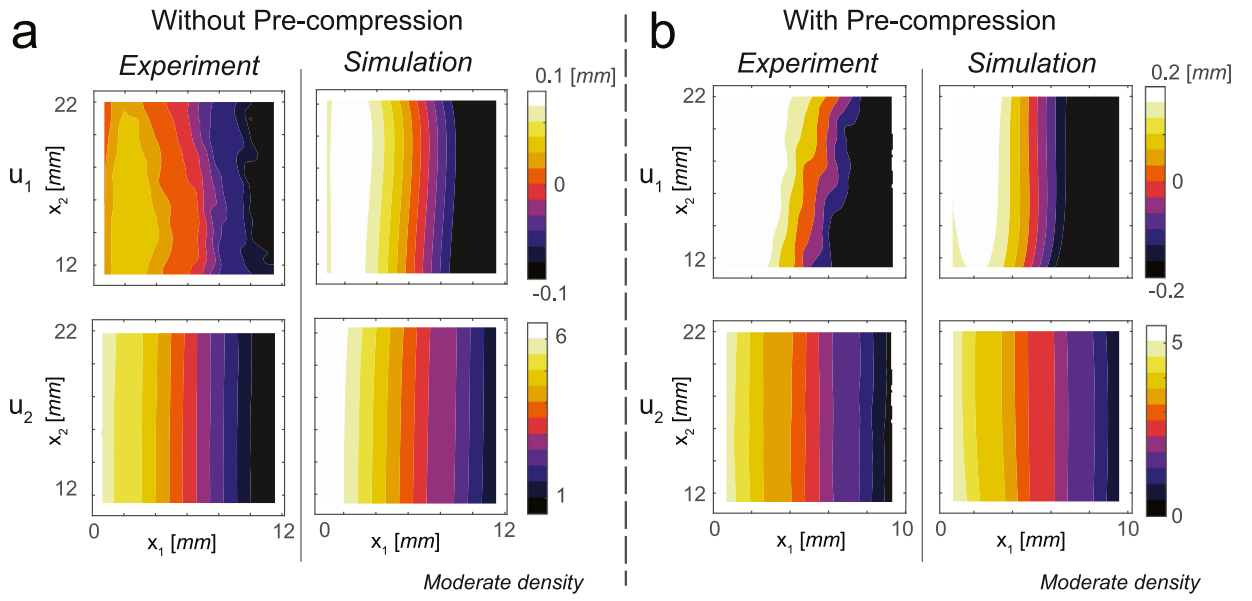


Fig. 8. Contours of the two-dimensional displacement field – i.e., $u_1(x_1, x_2)$ and $u_2(x_1, x_2)$ – on the front face of the specimen in the DIC region-of-interest from both experiment and simulation for simple-shear-like deformation (a) without pre-compression and at a shear displacement of $\delta/H = 0.5$ and (b) with pre-compression and at a shear displacement of $\delta/H = 0.375$ for the moderate density foam. For simple-shear-like deformation with pre-compression, the fields represent the displacement from the pre-compressed state to the deformed, sheared state.

plateau region is observed, and the slope of the force-displacement relation increases slightly as deformation progresses. Second, when pre-compression is applied, the force-displacement relation becomes more compliant, illustrating the interaction between volumetric and distortional aspects of the material response.

For the corresponding finite-element simulations, a fully three-dimensional model is utilized. The reference finite-element mesh, consisting of 97,200 Abaqus-C3D8H elements (eight-node, three-dimensional, hexahedral, constant-pressure elements) is shown in Fig. 7(a). Regarding boundary conditions, for simple-shear-like deformation *without* pre-compression, the right face of the mesh in Fig. 7(a) represents the fixed backer, on which all displacement components are prescribed to be zero. The left face of the mesh represents the moved backer, on which the vertical displacement u_2 is prescribed, while the other displacement components are constrained – i.e., $u_1 = 0$, $u_2 = \delta$, and $u_3 = 0$. The remaining faces are traction-free. The deformed mesh after a shear displacement of $\delta/H = 0.5$ is shown in Fig. 7(a), which upon close inspection, reveals the inhomogeneous nature of deformation near the boundaries in the simulation. For simple-shear-like deformation *with* pre-compression, pre-compression is applied by prescribing $u_1 = -0.25H$ and $u_2 = u_3 = 0$ on the right face of the mesh, while holding all displacement components at zero on the left face of the mesh. The deformed mesh after pre-compression is shown in Fig. 7(b), displaying some bulging in the vicinity of the traction-free faces. Then, a shear displacement is applied by fixing $u_1 = -0.25H$ and $u_2 = u_3 = 0$ on the right face of the mesh and prescribing $u_2 = \delta$ and $u_1 = u_3 = 0$ on the left face of the mesh. The deformed mesh after a shear displacement of $\delta/H = 0.375$ is shown in Fig. 7(b), displaying significant inhomogeneous deformation. The calculated shear force P versus shear displacement δ relations are included as dashed lines in Fig. 7(c) for each density both without and with pre-compression. For all densities, the model captures the features of the response both without and with pre-compression. In particular, it predicts both the lack of a plateau region in the response during shear-dominated deformation and the increased compliance with pre-compression. The ability of the model to capture the effect of pre-compression on the response in simple-shear-like deformation provides a partial validation of the coupled volumetric/distortional contribution to the free-energy function (3.2).

In addition to P versus δ relations, we also compare experimentally-measured and numerically-simulated displacement fields on the front face of the specimen. The DIC regions-of-interest are outlined with dashed lines in the reference, undeformed state and the deformed state at the maximum level of shear displacement for simple-shear-like deformation without and with pre-compression in Fig. 7(a) and (b), respectively, and contours of the two-dimensional, DIC-calculated displacement field – i.e., $u_1(x_1, x_2)$ and $u_2(x_1, x_2)$ – on the front face of the specimen in a representative experiment on the moderate density foam are shown in the left columns of Fig. 8(a) and (b). While in Fig. 8(a), the fields represent the displacement from the reference state to the deformed state, the fields in Fig. 8(b) for simple-shear-like deformation with pre-compression represent the displacement from the pre-compressed state to the deformed, sheared state.[10] The corresponding simulated

[10] This exception in the definition of the displacement field is only made in Fig. 8(b), and for notational economy, we continue to denote the displacement components as u_1 and u_2.

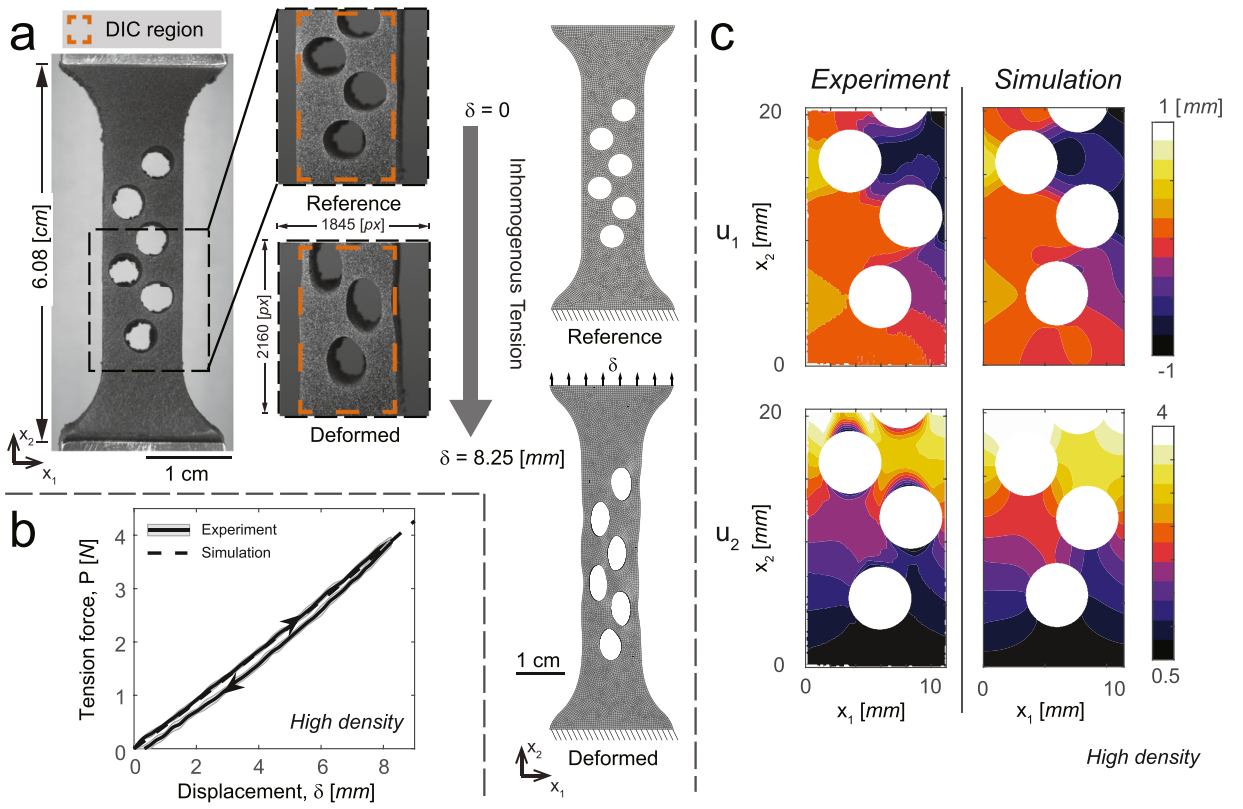


Fig. 9. A tension-dominated validation test for the high density foam consisting of tension of a specimen with circular holes, in which a standard dogbone geometry is modified with six asymmetrically-placed 2.5 mm radius holes. (a) Snapshot of the experimental specimen prior to deformation and the experimental region-of-interest used for DIC imaging in both the reference and the deformed states (DIC region approximately 1100×2000 px^2) as well as the corresponding finite-element configurations. (b) The top-grip force P versus displacement δ, in which solid lines with shaded error regions represent experimental data, and dashed lines are simulated predictions of the constitutive model. (c) Contours of the two-dimensional displacement field – i.e., $u_1(x_1, x_2)$ and $u_2(x_1, x_2)$ – on the front face of the specimen in the DIC region-of-interest from both experiment and simulation.

displacement fields are shown in the right columns of Fig. 8(a) and (b), demonstrating that the salient features of the experimental displacement fields are captured by the model. In particular, inhomogeneous deformation occurring due to edge-effects are clearly apparent in the u_1-fields in both experiments and simulations. Overall, the magnitudes and distributions of both the u_1- and u_2-components of the displacement field are consistent between experiment and simulation.

4.3. Tension of a specimen with circular holes

Finally, we conduct a validation test to evaluate the predictive capability of the constitutive model in a tension-dominated setting. Given the favorable comparisons of model predictions and experiments across all three densities in the cases of indentation and simple-shear-like deformation, we focus only on the high density foam in this final test. To obtain inhomogeneous deformation fields, we take inspiration from the work of Wang and Chester (2018) and cut circular holes in our tension specimens. Specifically, we begin with the dogbone geometry used in the simple tension characterization experiments described in Section 2 and laser-cut six, non-overlapping circular holes, each with a radius of 2.5 mm, at arbitrarily-chosen locations in the gage-section. This creates a purposefully asymmetric geometry, which is shown in Fig. 9(a). These modified specimens are then subjected to a tension-testing procedure analogous to that described in Section 2 for homogeneous simple tension experiments. The bottom grip is fixed, and the top grip moves upward at a displacement rate of $\dot{\delta} = 2.0 \times 10^{-3}$ mm/s. The measured tension force P as a function of the top grip displacement δ is shown in Fig. 9(b) as a solid line. As before, the shaded area denotes the range of one standard deviation above and below the average experimental measurement, and the arrows denote loading and unloading, demonstrating minimal hysteresis.

In the corresponding finite-element simulations, we model only the region between the grips, consisting of the gage-section and the section of the specimen between the grips and the gage-section. The specimen is modeled in three dimensions, matching the experimental geometry, and meshed with 185,216 Abaqus-C3D8H three-dimensional elements, as shown in Fig. 9(a). The front and back faces of the mesh as well as the side faces and the interior faces of the holes are traction-free. Regarding displacement boundary conditions, the bottom of the mesh is fully constrained – i.e., $u_1 = u_2 = u_3 = 0$ – and on the top surface of the mesh the vertical displacement u_2 is prescribed, while the other displacement components are

constrained – i.e., $u_1 = 0$, $u_2 = \delta$, and $u_3 = 0$. Fig. 6(b) shows the numerically-predicted force-displacement response as a dashed line, showing excellent agreement with the experimentally-measured relation.

As in the simple-shear-like validation case, we also compare experimentally-measured and numerically-simulated displacement fields on the front face of the specimen. The DIC region-of-interest is shown both in the reference state as well as the deformed state after a top-grip displacement of $\delta = 8.25$ mm in Fig. 9(a), outlined with dashed lines, and contours of the two-dimensional, DIC-calculated displacement field – i.e., $u_1(x_1, x_2)$ and $u_2(x_1, x_2)$ – on the front face of the specimen in a representative experiment are plotted in Fig. 9(c). The corresponding simulated front-face displacement fields are plotted alongside the experimental results in Fig. 9(c). There is strong quantitative agreement between the experimentally-measured and simulated displacement fields throughout the region-of-interest. The notable differences, specifically, the displacement gradients near the tops and bottoms of the holes in the experimental results, arise as an artifact of incorporating internal material-void boundaries within an Eulerian framework for the hybrid, cumulative-iterative displacement summation scheme used in the qDIC algorithm (see Landauer et al., 2018) and subsequent plotting in the reference configuration.

5. Discussion

5.1. Stability

A common drawback of stretch-based, compressible hyperelastic models (Ogden, 1972a; Storakers, 1986) is unstable behavior that limits the deformation range over which the model may be robustly applied. In this section, we examine the stability of the proposed hyperelastic model and illustrate the limits of stable behavior. Here, we assess stability by testing the ellipticity of the quasi-static governing equations – i.e., testing that the eigenvalues of the acoustic tensor in the deformed configuration remain positive for given homogeneous base states of deformation (e.g., Bigoni, 2012; Hill, 1962; Thomas, 1961; Triantafyllidis and Aifantis, 1986). For a homogeneous base state of deformation, the spatial acoustic tensor for a hyperelastic material is

$$Q_{ik}(\mathbf{n}) = \mathbb{C}_{ijkl} n_j n_l \quad \text{with} \quad \mathbb{C}_{ijkl} = J^{-1} F_{jn} F_{lq} \frac{\partial^2 \psi}{\partial F_{in} \partial F_{kq}}, \tag{5.1}$$

where $\mathbb{C}_{ijkl}$ are the spatial moduli, evaluated at the base state, and $\mathbf{n}$ is a unit vector. Details of the steps leading to (5.1) as well as an expression for the spatial moduli $\mathbb{C}_{ijkl}$ that is appropriate for use with a logarithmic-strain-based free-energy function are given in Appendix B. If the eigenvalues of $\mathbf{Q}(\mathbf{n})$ for a given base state of deformation remain positive for all unit vectors $\mathbf{n}$, the ellipticity of the governing equations is maintained, and homogeneous deformation is stable. However, if an eigenvalue of $\mathbf{Q}(\mathbf{n})$ is negative for any unit vector $\mathbf{n}$, the localization of deformation into bands becomes possible. To test for stability, we consider base states of deformation described by a spatial logarithmic strain tensor, $\mathbf{E}$, with invariants $\{K_1, K_2, K_3\}$ and principal directions $\{\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_3\}$ and all unit vectors $\mathbf{n}$ in three dimensions – i.e., $\mathbf{n} = \cos\theta \sin\gamma \mathbf{l}_1 + \sin\theta \sin\gamma \mathbf{l}_2 + \cos\gamma \mathbf{l}_3$ over the ranges $0 \le \theta \le 2\pi$ and $0 \le \gamma \le \pi$. In this way, we may identify whether unstable behavior is possible for a given combination of invariants $\{K_1, K_2, K_3\}$.[11]

Representative stability maps, generated using the free-energy density function introduced in Section 3 and the material parameters for the moderate density foam from Table 1, are shown in Fig. 10. Three specific modes of loading are considered in Fig. 10: (a) $K_3 = 1$ (simple tension), (b) $K_3 = -1$ (simple compression), and (c) $K_3 = 0$ (simple shear). Stable combinations of $\{K_1, K_2, K_3\}$ are denoted as white regions, while combinations of $\{K_1, K_2, K_3\}$ in which at least one eigenvalue of $\mathbf{Q}(\mathbf{n})$ is negative for at least one unit vector $\mathbf{n}$ are denoted as gray regions. In Fig. 10(a) and (b), the deformation paths for simple tension and simple compression are shown as solid lines, and in Fig. 10(c), an idealized homogeneous loading path for simple shear, representative of the inhomogeneous simple-shear-like deformation of Section 4.2, is shown. In all cases, the deformation paths traverse stable states.

While large regions of stable behavior are evident in Fig. 10, unstable behavior is also observed. The sources of unstable behavior may be linked to features of the model. First, the unstable regions in the bottom left corners of Fig. 10(a)–(c) arise because $X(K_1)$ is negative in this range when the material parameters for the moderate density foam are utilized. The phenomenological form for $X(K_1)$ (3.5) is an increasing function of K_1, which is enforced to be positive over the fitted range; however, for sufficiently negative values of K_1, $X(K_1) < 0$, leading to unstable behavior. If necessary, the functional form (3.5) may be extended to be positive for all K_1 to increase stable behavior. Second, since $dX/dK_1 > 0$ for all K_1, the shear-induced mean stress, $G_0(dX/dK_1)K_2^2$, is positive and may be of arbitrarily large magnitude as K_2 increases, leading to unstable behavior. In particular, the unstable region in the upper right domain of Fig. 10(b) is due to this effect. This feature is due to the low-order K_2-dependence of the coupled volumetric/distortional response and cannot be remedied by modifying the forms of the fitting functions. Instead, it would require an alternate structure for the decomposition of the free-energy function (3.2). Finally, unstable behavior for high values of K_2 can arise due to the higher-order distortional response function, $L(K_2, K_3)$, and its dependence on K_3. In particular, the unstable behavior in the top region of Fig. 10(c) is due to this effect. The functional form (3.6) may be extended with higher-order K_2-terms – perhaps motivated by locking

[11] We also utilize the analytical necessary and sufficient conditions for strong ellipticity of Dacorogna (2001) as an independent verification of the results presented in this section.

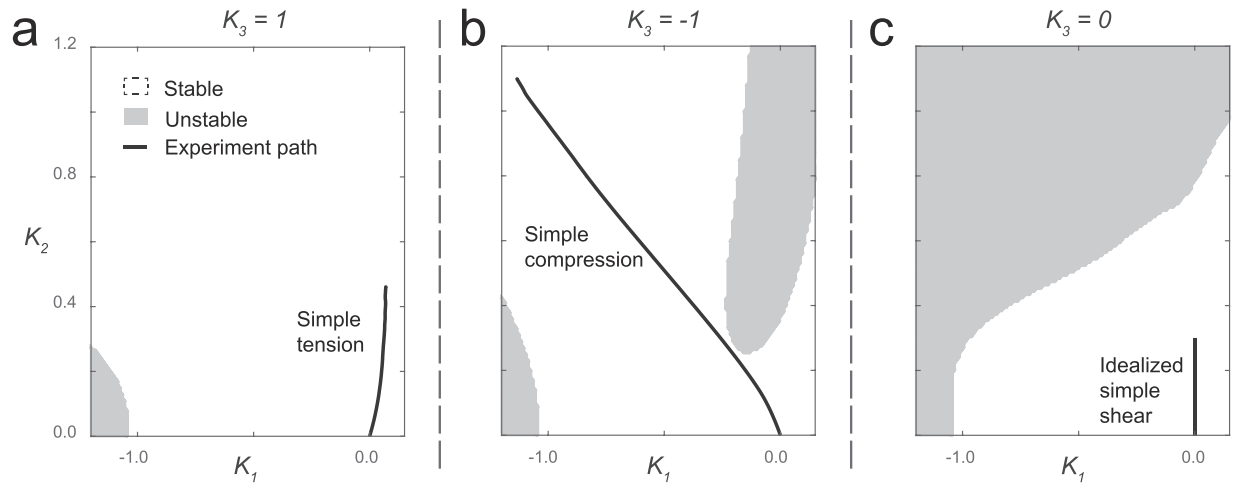


Fig. 10. Stability diagrams calculated using a loss-of-ellipticity test as a function of the invariants of the logarithmic strain K_1 and K_2 for several cases of fixed K_3: (a) $K_3 = 1$ (simple tension), (b) $K_3 = -1$ (simple compression), and (c) $K_3 = 0$ (simple shear). The white regions indicate stable behavior, while gray regions indicate the potential for unstable behavior. Solid lines correspond to representative deformation paths from simple tension, simple compression, and idealized simple shear, respectively.

behavior in the distortional response – to suppress unstable behavior for large values of K_2. In sum, the model exhibits robust, largely-stable behavior as evidenced by Fig. 10 as well as the numerical solutions obtained across a diverse set of inhomogeneous deformation scenarios in Section 4.

5.2. Comparison to other models

In this section, we compare the ability of the proposed model to capture experimental data with that of other important hyperelastic models for porous elastomers in the literature. We begin with the classic model of Blatz and Ko (1962) – a phenomenological, invariant-based free-energy function for porous elastomers – which depends on the second and third principal invariants of $\mathbf{C}$ as follows:

$$\psi = \frac{G_0}{2}\left(\frac{I_2}{I_3} + 2\sqrt{I_3} - 5\right), \tag{5.2}$$

where G_0 is the ground-state shear modulus of the foam. We note that (5.2) represents a simplified version of a broader three-parameter model and implies a ground-state Poisson's ratio of 0.25. Since the ground-state Poisson's ratio of the moderate density Poron XRD foam is about 0.27, we utilize the simplified form (5.2) along with the previously-fitted value of $G_0 = 65.2$ kPa (Table 1). The resulting engineering stress versus axial strain and lateral strain versus axial strain response in simple compression/tension for the Blatz-Ko model is shown in Fig. 11(a) and (b) using square (□) markers along with the experimental data for the moderate density foam. As expected for the highly-porous foam under consideration, the Blatz-Ko model does a reasonable job capturing data in the low to moderate strain regime, i.e., axial engineering strains between approximately −0.1 and 0.2, but is not applicable beyond this regime.

Next, we consider two homogenization-based models for porous elastomers with spherical voids: the model of Danielsson et al. (2004) and the more recent model of Shrimali et al. (2019). Both models consider microstructures consisting of closed-cell voids and do not account for buckling in the elastomeric-matrix microstructure during compression. Therefore, they are limited in scope to low to moderate porosity foams, and we do not expect these models to capture experimental data for the highly-porous, open-cell foam under consideration. However, since these models are two of only a handful of explicit, compressible hyperelasticity models, we include them in our discussion for the sake of comparison. First, the approach of Danielsson et al. (2004), when specialized to the case of a neo-Hookean matrix, yields the following free-energy function that depends on the invariants $I_1 = \mathrm{tr}\,\mathbf{C}$ and $J = \det\mathbf{F}$:

$$\psi = \frac{G_\mathrm{m}}{2}\left[I_1\left(2 - \frac{1}{J} - \frac{\phi_0 + 2(J-1)}{J^{2/3}(1+(J-1)/\phi_0)^{1/3}}\right) - 3(1-\phi_0)\right], \tag{5.3}$$

where G_m is the ground-state shear modulus of the matrix material and ϕ_0 is the void volume fraction of the undeformed porous material. The ground-state shear modulus of the foam material is then $G_0 = G_\mathrm{m}(1-\phi_0)$. From Table 1, we take $G_0 = 65.2$ kPa and estimate $\phi_0 = 1 - J_\mathrm{min} = 0.84$, so that $G_\mathrm{m} = 407.5$ kPa. The corresponding engineering stress versus axial strain and lateral strain versus axial strain response in simple compression/tension is shown in Fig. 11(a) and (b) using triangle (△) markers. We note that $\phi_0 = 1 - J_\mathrm{min}$ is a rough estimate of the initial void volume fraction. For this reason, we have considered values of ϕ_0 over the range $0.75 \le \phi_0 \le 0.95$, i.e., the typical range for the low, moderate, and high

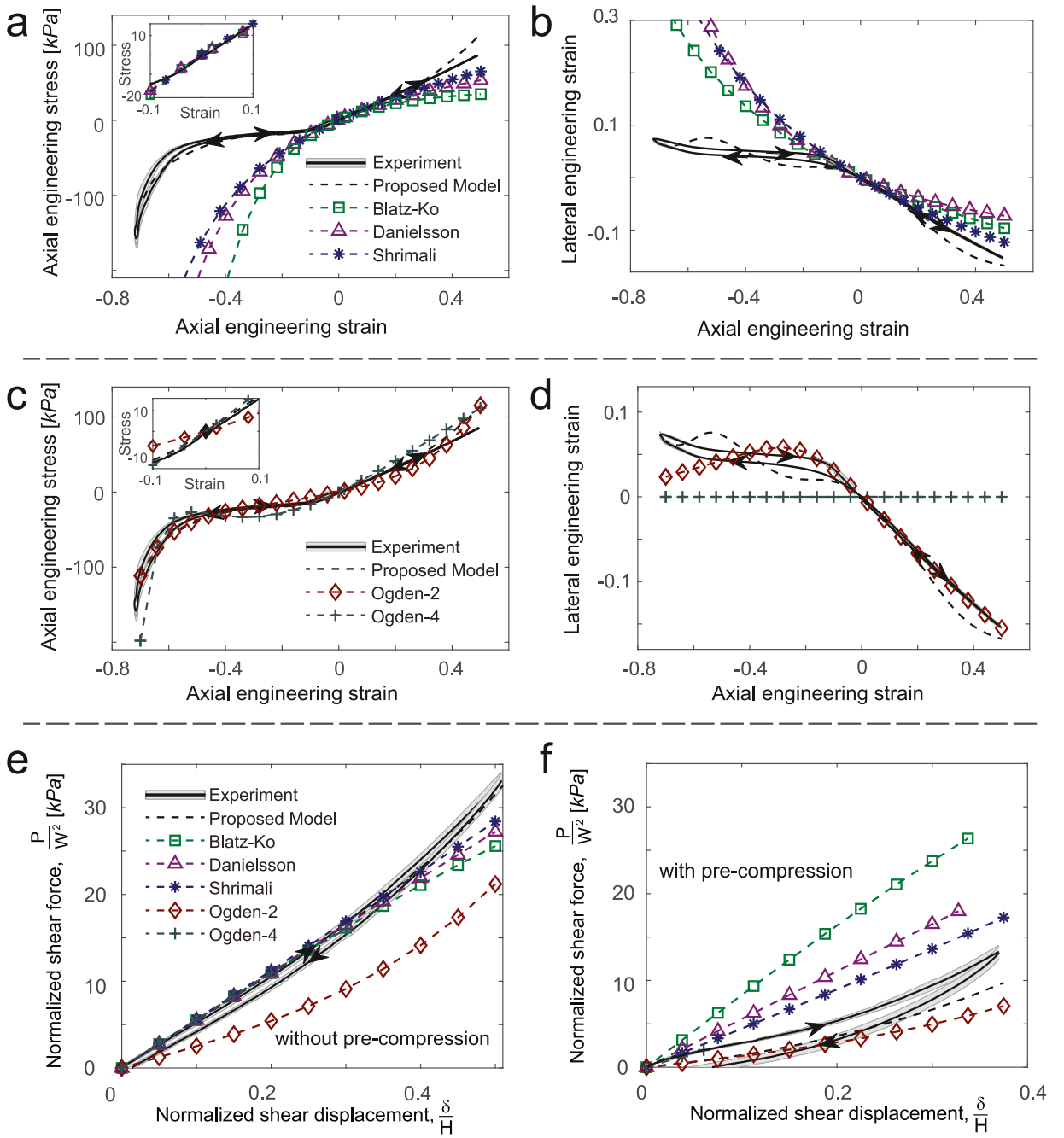


Fig. 11. Comparison of fits obtained using hyperelastic models for porous elastomers in the literature with the proposed approach for the moderate density Poron XRD foam. (a) Axial engineering stress and (b) lateral engineering strain versus axial engineering strain fits for the proposed model, the Blatz and Ko (1962) model (Blatz-Ko), the Danielsson et al. (2004) model (Danielsson), and the Shrimali et al. (2019) model (Shrimali). (c) Axial engineering stress and (d) lateral engineering strain versus axial engineering strain fits for a two-term Ogden-Storakers fit (Ogden-2) and a four-term Ogden-Storakers fit (Ogden-4) (Ogden, 1972a; Storakers, 1986). (*Insets (a) and (c)*) Close-up of the moderate-strain regime. Normalized shear force P/W^2 versus normalized shear displacement δ/H for simple-shear-like deformation both (e) without and (f) with pre-compression for all models considered as well as experimental data for the moderate density Poron XRD foam.

density Poron XRD foams, with $G_0 = 65.2$ kPa fixed and confirmed that within this range, the precise value of ϕ_0 has a minimal impact on the response shown in Fig. 11(a) and (b). Second, the homogenization-based free-energy function of Shrimali et al. (2019) – also specialized to the case of a neo-Hookean matrix – is

$$\psi = \frac{3G_m(1-\phi_0)}{2(3+2\phi_0)}(I_1-3) + \frac{3G_m}{2J^{1/3}}\left[2J - 1 - \frac{(1-\phi_0)J^{1/3}(3J^{2/3}+2\phi_0)}{3+2\phi_0} - \frac{\phi_0^{1/3}J^{1/3}(2J+\phi_0-2)}{(J-1+\phi_0)^{1/3}}\right], \tag{5.4}$$

where G_m and ϕ_0 continue to denote the ground-state shear modulus of the matrix material and the undeformed void volume fraction. The ground-state shear modulus of the foam material for this model is $G_0 = 3G_m(1-\phi_0)/(3+2\phi_0)$, and again, we take $G_0 = 65.2$ kPa and $\phi_0 = 0.84$, so that $G_m = 635.7$ kPa for this case. The corresponding response is included in Fig. 11(a) and (b) using star (*) markers. As for the model of Danielsson et al. (2004), we have confirmed that the response shown in Fig. 11(a) and (b) for the model of Shrimali et al. (2019) is minimally affected as the value of ϕ_0 is varied over the range $0.75 \leq \phi_0 \leq 0.95$, while $G_0 = 65.2$ kPa remains fixed. As expected, both models yield similar predictions and are capable of capturing data in the moderate strain regime but do not capture the response beyond this regime – especially the response in the plateau regime in compression.

Finally, we consider the stretch-based Ogden-Storakers model (Ogden, 1972a; Storakers, 1986). This is the most commonly utilized phenomenological hyperelasticity model for highly-compressible foams and is built in to Abaqus (2018) as the "hyperfoam" material. The free-energy function for the Ogden-Storakers model is given through the principal stretches $\{\lambda_1, \lambda_2, \lambda_3\}$ and $J = \lambda_1\lambda_2\lambda_3$ as

$$\psi = \sum_{n=1}^{N} \frac{2G_n}{\alpha_n^2}\left[\lambda_1^{\alpha_n} + \lambda_2^{\alpha_n} + \lambda_3^{\alpha_n} - 3 + \frac{1}{\beta_n}\left(J^{-\alpha_n\beta_n} - 1\right)\right], \tag{5.5}$$

where N is the number terms and $\{G_n, \alpha_n, \beta_n\}$ are material properties for each term. We consider two-term and four-term fits, which we refer to as Ogden-2 and Ogden-4, respectively. For both cases, we utilize the material parameter fitting routines built in to Abaqus. The six parameters of the Ogden-2 fit were determined to be $\{G_1 = 24.4\,\text{kPa}, \alpha_1 = 8.31, \beta_1 = -0.154, G_2 = 4.18\,\text{kPa}, \alpha_2 = -1.39, \beta_2 = 7.71\}$, and the corresponding fitted response is shown in Fig. 11(c) and (d) using diamond (◊) markers. Although the Ogden-2 fit captures the overall shape of the axial stress and lateral strain versus axial strain responses, the accuracy of the fit in the moderate-strain regime is sacrificed, as shown in the inset of Fig. 11(c). Regarding the Ogden-4 fit, we found that to enable the nonlinear fitting routine in Abaqus to converge, the lateral strain response must be fixed to zero, so that all $\beta_n = 0$ (see Fig. 11(d)). Hence, there are eight fitted parameters in total for the Ogden-4 fit, which were determined to be $\{G_1 = 568\,\text{kPa}, \alpha_1 = 0.867, G_2 = 610\,\text{kPa}, \alpha_2 = 0.859, G_3 = -1.71\,\text{MPa}, \alpha_3 = 0.165, G_4 = 627\,\text{kPa}, \alpha_4 = -0.477\}$. Although the axial stress versus axial strain response of the Ogden-4 fit – shown in Fig. 11(c) using plus (+) markers – more closely matches experimental data in the moderate-strain regime compared to the Ogden-2 fit, the response is non-monotonic in compression, which suggests unstable behavior. Indeed, the loss-of-ellipticity test reveals unstable behavior for a large set of deformation states for the Ogden-4 fit. For this class of highly-compressible, open-cell elastomeric foams, it is typically quite difficult to obtain a fit using stretch-based models that captures both the axial stress and lateral strain versus axial strain responses and behaves stably over a wide range of deformation states, and the inclusion of additional terms does not change this observation. For example, we have considered five-term and six-term Ogden-type fits, while taking all $\beta_n = 0$ so that the fits involve ten and twelve parameters, respectively, and we observe unstable behavior similar to that of the Ogden-4 fit.

We also compare predictions of all models considered in this section against the experimental data from inhomogeneous simple-shear-like deformation both without and with pre-compression described in Section 4.2. Model predictions are obtained using finite-element simulations, and the calculated shear force P versus shear displacement δ relations are shown in Fig. 11(e) and (f) along with experimental data for the moderate density foam. The Blatz-Ko model (□) does not capture the increased compliance in shear with pre-compression, and while both homogenization-based models (△ and *) do predict some increase in compliance with pre-compression, they do not quantitatively capture the normalized force versus displacement response in simple-shear-like deformation with pre-compression. The Ogden-2 fit (◊) does not capture the experimental data for simple-shear-like deformation without pre-compression but is consistent with the data for simple-shear-like deformation with pre-compression. Finally, due to instabilities in the fitted model, numerical solutions using the Ogden-4 fit (+) cannot be obtained beyond moderate shear displacements – namely, $\delta/H = 0.3$ and $\delta/H = 0.08$ for simple-shear-like deformation without and with pre-compression, respectively. These results illustrate the difficulties in robustly capturing experimental data for highly-compressible, open-cell elastomeric foams and the improved predictions enabled by the proposed approach.

6. Concluding remarks

Summary. In this work, we presented an extensive new set of experiments on three densities of the polyurethane-based, open-cell elastomeric foam "Poron XRD". Experiments consisted of homogeneous simple compression/tension as well as three types of inhomogeneous experiments: spherical and conical indentation, simple-shear-like deformation without and with pre-compression, and tension of a specimen with circular holes. A phenomenological, isotropic, finite-deformation, hyperelastic constitutive model based on invariants of the logarithmic, Hencky strain was proposed. A key simplifying assumption of our modeling approach is that volumetric/distortional coupling only involves low-order dependence on the magnitude of distortional deformation – which leads to a more straightforward interpretation of the fitting functions and a systematic path for material parameter estimation from simple compression/tension data (see Appendix A). In compression and tension, the fitted model for each of the three densities of Poron XRD faithfully captures the nonlinear stress versus axial strain and lateral strain versus axial strain responses, in particular, the tension/compression asymmetry featuring a nearly-flat plateau regime in compression. The constitutive model was implemented in Abaqus (2018) using a user-material

subroutine, which was used to obtain model predictions in inhomogeneous deformation settings for the purpose of validation. The model predictions were shown to be consistent with experimental data in the inhomogenous validation cases across compression, shear, and tension-dominated settings. All simple compression/tension experimental data and resulting fitted model data for Poron XRD has been made available to the community via GitHub (https://github.com/FranckLab) and as attached research data. The user-material subroutine implementation of the constitutive model and sample input files for several of the validation cases are also made available (https://github.com/HenannResearchGroup). Additional information is freely available upon request.

Limitations. Our hyperelastic model was developed to capture the equilibrium response of open-cell, isotropic elastomeric foams with relatively high void volume fraction. As such, the model has several operational limitations arising from assumptions made in model development. First, many elastomeric foam materials exhibit strain-rate-dependent, viscoelastic material behavior, whereas the proposed model is rate-independent. Second, the polymeric matrix material of the foam may be temperature sensitive, in some cases dramatically so, and this effect may be coupled with the rate-sensitivity of the matrix material. All experiments in the present work were conducted under approximately isothermal conditions at room temperature. Accordingly, the model also assumes isothermal conditions, and the effect of variable temperature is not captured. To use the model at elevated or reduced temperature, a new set of experiments at the temperature of interest would be required. Third, depending upon the microstructure, elastomeric foam materials may exhibit strain localization into compaction bands under uniform external loading. The foam material considered in the present work did not exhibit localization, and localization behavior is not accounted for in the model.

Future work. There are numerous avenues for future work. This study focused on a single polyurethane-based, open-cell elastomeric foam material over a moderate range of densities. We expect that the mechanical behavior of other open-cell foam materials may be well captured by the proposed model, so long as the material does not exhibit discernible strain localization. Future work will apply the proposed modeling approach to other elastomeric foam materials. We expect that improvements will be made to the phenomenological fitting functions utilized in the present work, either driven by additional experimental data for different elastomeric foam materials or motivated by micromechanical arguments. Furthermore, while the model has been tested in compression, shear, and tension-dominated situations, model validation has not exhaustively spanned all combinations of the invariants $\{K_1, K_2, K_3\}$. For example, it remains to test the proposed model in equibiaxial tension, in which $K_1 > 0$ and $K_3 = -1$ (cf., e.g., the simple compression, $K_1 < 0$ and $K_3 = -1$, or simple tension, $K_1 > 0$ and $K_3 = 1$, modes used in model calibration). Beyond hyperelastic modeling of elastomeric foams, we envision that the proposed modeling framework may be extended to incorporate the effects of strain-rate and temperature sensitivity by adapting existing finite viscoelasticity frameworks for monolithic elastomers (e.g., Bergstrom and Boyce, 1998; Reese and Govindjee, 1998). Finally, localization arising from microstructural defects is a key aspect of deformation in very-low-density foams, and developing a model that captures this behavior will be the subject of future work.

Acknowledgments

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Appendix A. Material parameter estimation procedure

The constitutive model described in Section 3 involves 14 material parameters that are used to represent the hyperelastic behavior of non-localizing, open-cell elastomeric foams. This Appendix provides a heuristic procedure for estimating the values of these parameters for a generic elastomeric foam material. The procedure is illustrated by estimating the material parameters for the high density Poron XRD foam, using the experimental data from simple compression/tension experiments shown in Fig. 1(c).

The first step of the procedure is to determine the ground-state shear and bulk moduli, G_0 and B. The ground-state Young's modulus and Poisson's ratio are extracted from the experimental data via a linear least squares regression to the compression and tension experimental data in the axial engineering strain range of approximately -0.02 to 0.02 and used to calculate G_0 and B. For the high density foam, $G_0 = 102.0$ kPa and $B = 193.8$ kPa. Second, the phenomenological functions $X(K_1)$, $L(K_2, K_3)$, and $f(K_1)$ are iteratively fit to data extracted from the axial stress and lateral strain versus axial strain data in simple compression, in which $K_3 = -1$ is fixed. The parameters used to fit the compression response consist of $\{X_1', K_1^0, \Delta_K, X_2'\}$ for $X(K_1)$, $\{C_0, p\}$ for $L(K_2, K_3)$, and $\{J_{\min}, C_2, C_3, r\}$ for $f(K_1)$. Finally, the remaining parameters – namely, $\{C_1, q\}$ for $L(K_2, K_3)$ – are then estimated from the simple tension experimental data, in which $K_3 = 1$ is fixed. The processes for estimating the compression-based and tension-based parameters are discussed in the following sections, and Fig. A.1 outlines the steps in this workflow.

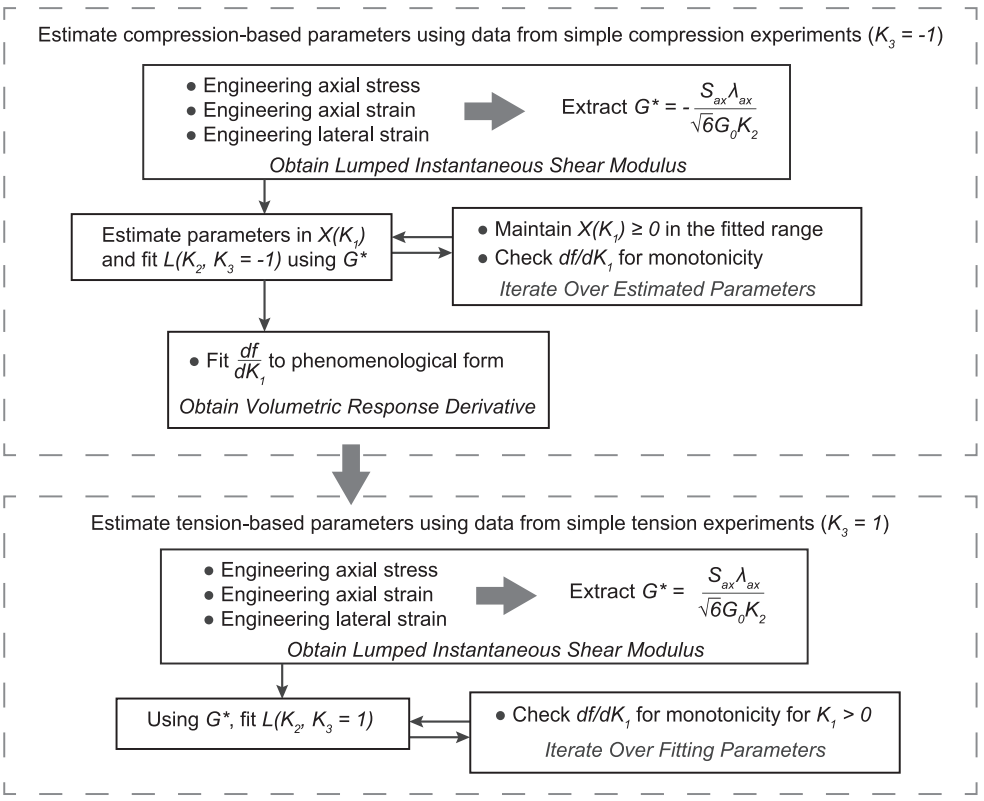


Fig. A.1. Outline of the procedure for estimating material parameters from experimental data for simple compression and tension. Compression-based parameters are estimated first, followed by the estimation of tension-based parameters.

A1. Estimation of compression-based parameters

In simple compression, the principal stretches are $\lambda_1 = \lambda_2 = \lambda_{\text{lat}} > 1$ and $\lambda_3 = \lambda_{\text{ax}} < 1$, and the logarithmic strain invariants are $K_1 = \ln(\lambda_{\text{ax}}\lambda_{\text{lat}}^2)$, $K_2 = (\sqrt{6}/3)\ln(\lambda_{\text{lat}}/\lambda_{\text{ax}})$, and $K_3 = -1$, so that the histories of K_1 and K_2 during compression may be determined from the kinematic measurements. The principal engineering stresses are $S_1 = S_2 = 0$ and $S_3 = S_{\text{ax}} < 0$. Using (3.3) and the definition of the first Piola stress, $\mathbf{S} = J\mathbf{T}\mathbf{F}^{-\top}$, the components of the constitutive equation for the stress are

$$\begin{aligned} S_{\text{ax}} &= \frac{1}{\lambda_{\text{ax}}}\left[\left(G_0\frac{dX}{dK_1}K_2^2 + B\frac{df}{dK_1}\right) - \frac{2}{\sqrt{6}}G_0\left(2X(K_1)K_2 + \left.\frac{\partial L}{\partial K_2}\right|_{K_3=-1}\right)\right], \\ 0 &= \frac{1}{\lambda_{\text{lat}}}\left[\left(G_0\frac{dX}{dK_1}K_2^2 + B\frac{df}{dK_1}\right) + \frac{1}{\sqrt{6}}G_0\left(2X(K_1)K_2 + \left.\frac{\partial L}{\partial K_2}\right|_{K_3=-1}\right)\right]. \end{aligned} \tag{A.1}$$

Next, combining $(A.1)_1$ and $(A.1)_2$ in order to eliminate df/dK_1 and rearranging, we obtain

$$G^* = X(K_1) + \frac{1}{2K_2}\left.\frac{\partial L}{\partial K_2}\right|_{K_3=-1} = -\frac{S_{\text{ax}}\lambda_{\text{ax}}}{\sqrt{6}G_0K_2}. \tag{A.2}$$

where G^* is a normalized lumped instantaneous shear modulus. All quantities on the right-hand-side of (A.2) – namely, the histories of S_{ax}, λ_{ax}, and K_2 during simple compression as well as the ground-state shear modulus G_0 – are known, so that (A.2) may be used to determine the history of G^* during simple compression, which may, in turn, be used to estimate the parameters appearing in $X(K_1)$ and $L(K_2, K_3)$. The measured G^* history during simple compression, projected onto the G^*-K_1-plane and the G^*-K_2-plane, is shown as solid lines in Fig. A.2(a) and (b), respectively. An iterative process is necessary to estimate the material parameters appearing in the $X(K_1)$ and $L(K_2, K_3 = -1)$ functions, while ensuring that – for the purpose of stability – a monotonically increasing fitting function for df/dK_1 may be used to capture data. A method for choosing an initial guess for the parameters appearing in $X(K_1)$ is as follows:

1. The parameter X_1' is the slope of $X(K_1)$ prior to the plateau regime in compression. The value of X_1' may be estimated from the slope of the projection of G^* versus K_1 at $K_1 = 0$, as illustrated in Fig. A.2(a).

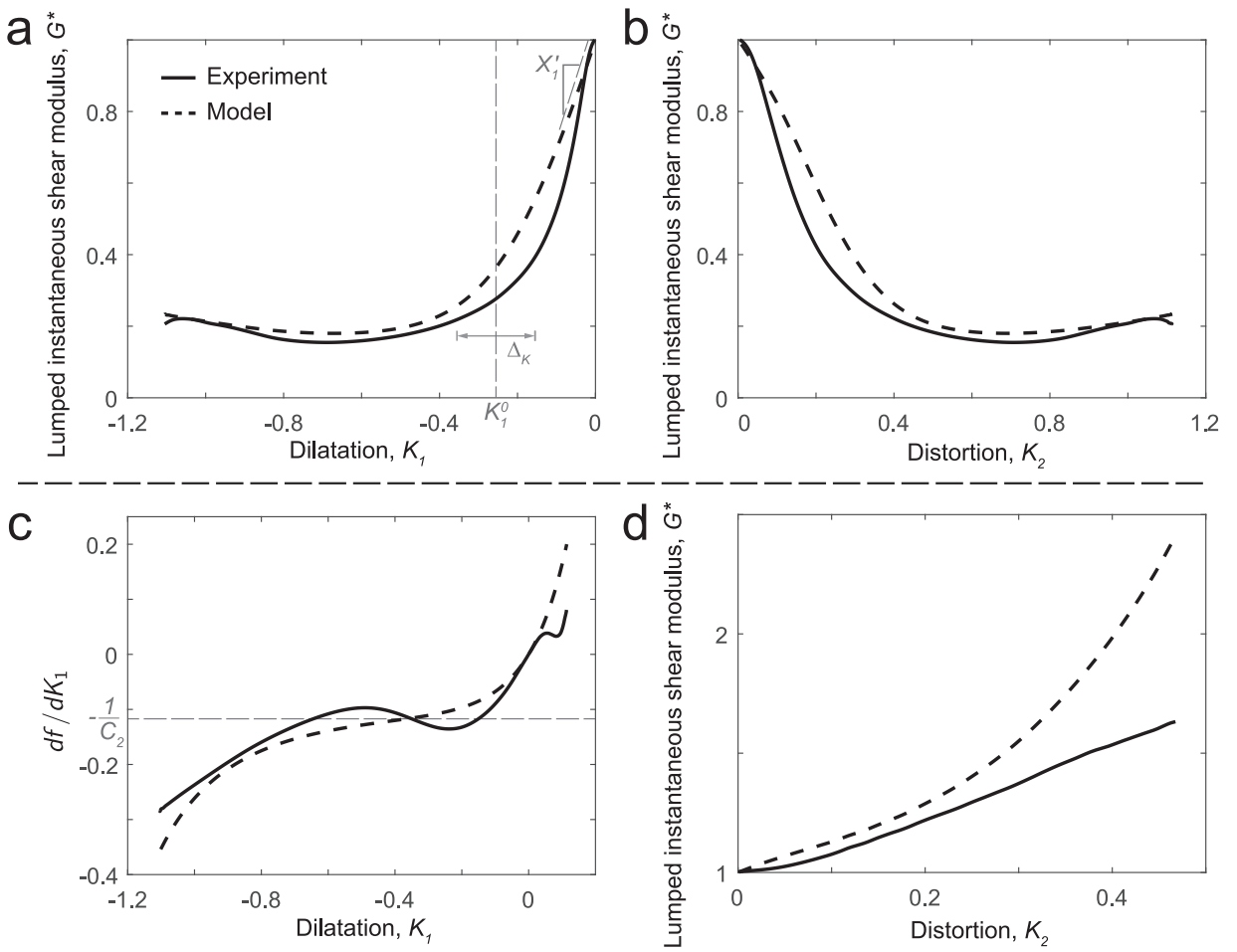


Fig. A.2. Projections of the normalized lumped instantaneous shear modulus, G^*, plotted as a function of (a) the dilatation, K_1, and (b) the distortion, K_2, for a simple compression loading history ($K_3 = -1$). (c) The first derivative of the volumetric response function $f(K_1)$ with respect to K_1, df/dK_1, plotted as a function of K_1. (d) Projection of the normalized lumped instantaneous shear modulus, G^*, plotted as a function of the distortion, K_2, for a simple tension loading history ($K_3 = 1$). Solid lines denote experimental data for the high density Poron XRD foam material; dashed lines denote the model fit; and interpretations of the material parameters X_1', K_1^0, Δ_K, and C_2 are illustrated.

2. The parameter K_1^0 is the value of K_1 denoting the transition to the plateau regime in $X(K_1)$. The value of K_1^0 may be estimated as the center of the K_1-range, in which the slope of G^* versus K_1 in the projection of Fig. A.2(a) changes rapidly during simple compression, as illustrated in Fig. A.2(a).
3. The parameter Δ_K is the width of the K_1-range across which the value of dX/dK_1 transitions from X_1' to X_2'. The value of Δ_K may be estimated from the width of the transition range in K_1 for G^* versus K_1 in the projection of Fig. A.2(a) during simple compression, as illustrated in Fig. A.2(a).
4. The parameter X_2' is the slope of $X(K_1)$ after the onset of the plateau regime in compression. The value of X_2' is estimated to be small compared to X_1' – i.e., $X_2' \sim O(X_1'/10)$. For the purpose of stability, it is important that the choice of this parameter results in $X(K_1) > 0$ over the entire fitted range of K_1.

Guided by Fig. A.2(a), we estimate the material parameters appearing in the $X(K_1)$ function to be $X_1' = 3.7$, $K_1^0 = -0.21$, $\Delta_K = 0.2$, and $X_2' = 0.22$ for the Poron XRD foam material.

Once the parameters of the $X(K_1)$ function are estimated, the next step is to estimate the parameters of the $L(K_2, K_3 = -1)$ function, namely, C_0 and p. Rearranging (A.2), we obtain

$$\left.\frac{\partial L}{\partial K_2}\right|_{K_3=-1} = 2K_2\left(-\frac{S_{ax}\lambda_{ax}}{\sqrt{6}G_0K_2} - X(K_1)\right). \tag{A.3}$$

Using the measured data and the previously-estimated parameters, (A.3) may be used to determine C_0 and p, which we have estimated to be $C_0 = 0.1$ and $p = 4$ for the Poron XRD foam material. Projections of the fitted G^* history during simple

compression, i.e., $G^* = X(K_1) + (1/(2K_2))\partial L/\partial K_2|_{K_3=-1}$, using the estimated material parameters are shown as dashed lines in Fig. A.2(a) and (b).

With initial estimates for X_1', K_1^0, Δ_K, X_2', C_0, and p in hand, we check that a monotonically increasing fitting function for df/dK_1 may be used to capture data. Rearranging $(A.1)_2$, we obtain the following expression for df/dK_1:

$$\frac{df}{dK_1} = -\frac{G_0}{B}\left[\frac{1}{\sqrt{6}}\left(2X(K_1)K_2 + \left.\frac{\partial L}{\partial K_2}\right|_{K_3=-1}\right) + \frac{dX}{dK_1}K_2^2\right]. \tag{A.4}$$

Again, all quantities on the right-hand-side of (A.4) are either measured data or previously-fitted parameters and are therefore known. The df/dK_1 history during simple compression (A.4) is shown as a solid line in Fig. A.2(c), using the previously-estimated parameters. This history is non-monotonic[12] but may be reasonably captured by a monotonically-increasing fitting function. If the df/dK_1 history is too non-monotonic, the material parameters of the $X(K_1)$ function should be adjusted. Typically, the parameter X_1' should be decreased from its initial estimated value, and the parameter X_2' should be increased – all while ensuring that $X(K_1) \geq 0$ over the fitted K_1-range. Then, the $L(K_2, K_3 = -1)$ function is refit, and the resulting df/dK_1 history may be checked again for near-monotonicity. In this way, the material parameters of the $X(K_1)$ function are iteratively adjusted until the df/dK_1 history may be reasonably captured by a monotonically-increasing fitting function. Once this is achieved, the parameters $J_{\min}$, C_2, C_3, and r may be estimated. The parameter $J_{\min}$ represents the minimum allowable value of J and may be interpreted as $J_{\min} = 1 - \phi_0$, where ϕ_0 is the void volume fraction. Therefore, $\ln J_{\min}$ is the minimum allowable value of K_1, and df/dK_1 diverges as $K_1 \to \ln J_{\min}$. The parameter C_2 sets the value of df/dK_1 in the plateau region. More specifically, the value of df/dK_1 in the plateau region is slightly more negative than $(-1/C_2)$, as illustrated in Fig. A.2(c). Finally, the parameters C_3 and r are adjustable parameters that may be chosen so that the model response matches the stiffening behavior after the plateau regime. The values of the parameters for the high density Poron XRD foam material, determined using this procedure, are $J_{\min} = 0.19$, $C_2 = 9$, $C_3 = 0.026$, and $r = 2$, and the fitted df/dK_1 function is plotted as a dashed line in Fig. A.2(c).[13]

A2. Estimation of tension-based parameters

Once the response in simple compression has been fit, experimental data from simple tension is used to estimate the remaining parameters in the $L(K_2, K_3)$ function, namely, C_1 and q. As in the material parameter estimation procedure described above for simple compression, estimating these parameters is an iterative process, since it must be ensured that df/dK_1 may be reasonably captured by a monotonically-increasing fitting function over the positive K_1-range of application. In simple tension, the principal stretches are $\lambda_1 = \lambda_{ax} > 1$ and $\lambda_2 = \lambda_3 = \lambda_{lat} < 1$; the logarithmic strain invariants are $K_1 = \ln(\lambda_{ax}\lambda_{lat}^2)$, $K_2 = (\sqrt{6}/3)\ln(\lambda_{ax}/\lambda_{lat})$, and $K_3 = 1$; and the principal engineering stresses are $S_1 = S_{ax} > 0$ and $S_2 = S_3 = 0$. The components of the constitutive equation for the stress are then

$$\begin{aligned} S_{ax} &= \frac{1}{\lambda_{ax}}\left[\left(G_0\frac{dX}{dK_1}K_2^2 + B\frac{df}{dK_1}\right) + \frac{2}{\sqrt{6}}G_0\left(2X(K_1)K_2 + \left.\frac{\partial L}{\partial K_2}\right|_{K_3=1}\right)\right], \\ 0 &= \frac{1}{\lambda_{lat}}\left[\left(G_0\frac{dX}{dK_1}K_2^2 + B\frac{df}{dK_1}\right) - \frac{1}{\sqrt{6}}G_0\left(2X(K_1)K_2 + \left.\frac{\partial L}{\partial K_2}\right|_{K_3=1}\right)\right], \end{aligned} \tag{A.5}$$

and combining $(A.5)_1$ and $(A.5)_2$ in order to eliminate df/dK_1 and rearranging, we obtain the normalized lumped instantaneous shear modulus, G^*, in tension:

$$G^* = X(K_1) + \frac{1}{2K_2}\left.\frac{\partial L}{\partial K_2}\right|_{K_3=1} = \frac{S_{ax}\lambda_{ax}}{\sqrt{6}G_0K_2}, \tag{A.6}$$

[12] We note that the non-monotonic behavior in the df/dK_1 history of Fig. A.2(c) arises due to the presence of the shear-induced mean stress term, $(dX/dK_1)K_2^2$, in (A.4).

[13] We note that the material parameter estimation procedure described in this section ensures that the fitted engineering stress versus axial engineering strain response remains monotonic in simple compression in order to maximize stable behavior. However, the model fit to the lateral engineering strain versus axial engineering strain response can be slightly non-monotonic (see Fig. 1). Since non-monotonicity in the lateral strain response does not affect stability, and since the oscillations in the lateral strain response are on the order of 0.01 whereas the axial strain varies over an order of 0.1, we do not prioritize the elimination of the relatively low amplitude oscillations in the fitted lateral strain response in the present work.

which may be rearranged as

$$\left.\frac{\partial L}{\partial K_2}\right|_{K_3=1} = 2K_2\left(\frac{S_{\text{ax}}\lambda_{\text{ax}}}{\sqrt{6}G_0K_2} - X(K_1)\right). \tag{A.7}$$

Using the measured data and previously-estimated parameters, (A.7) may be used to estimate the remaining two parameters appearing in the $L(K_2, K_3 = 1)$ function, C_1 and q. Then, rearranging $(A.5)_2$, we obtain the following expression for df/dK_1 during simple tension:

$$\frac{df}{dK_1} = \frac{G_0}{B}\left[\frac{1}{\sqrt{6}}\left(2X(K_1)K_2 + \left.\frac{\partial L}{\partial K_2}\right|_{K_3=1}\right) - \frac{dX}{dK_1}K_2^2\right]. \tag{A.8}$$

Using the estimated values for the parameters C_1 and q, (A.8) is used to check that the df/dK_1 history is nearly-monotonic for $K_1 > 0$. If this is not the case, the value of C_1 should be increased from its initial estimated value. If it is necessary to substantially increase C_1 to obtain a monotonic df/dK_1 history for $K_1 > 0$, so that the G^* history from simple tension is significantly overestimated, the estimation of the compression-based parameters should be repeated using a reduced value for X_1', and in this way, the parameters are iteratively adjusted until the df/dK_1 history is nearly monotonic. Using this process, we estimate the parameters for the high density Poron XRD foam material to be $C_1 = 1.9$ and $q = 5$, and the resulting fitted G^* history during simple tension, projected in the G^*-K_2-plane, is shown in Fig. A.2(d) as a dashed line, along with the measured G^* history as a solid line. It is evident that a compromise is made in which G^* is overestimated in simple tension in order to obtain a suitable df/dK_1 history.

Appendix B. Loss of ellipticity

In this appendix, we provide background for the loss of ellipticity analysis utilized in Section 5.1. We consider a time-dependent, incremental process superimposed upon a time-independent base state of deformation (Ogden, 1984). The base state is described by a motion field $\boldsymbol{\chi}(\mathbf{x})$, which satisfies the governing equations. Then, denoting small increments with a superposed dot, the incremental displacement field is expressed referentially as $\dot{\mathbf{u}}(\mathbf{x}, t)$ or, alternatively, expressed spatially as $\dot{\mathbf{u}}(\boldsymbol{\chi}^{-1}(\mathbf{y}), t)$. The associated incremental deformation gradient and incremental spatial displacement gradient fields are $\dot{\mathbf{F}} = \nabla\dot{\mathbf{u}}$ and $\dot{\mathbf{H}} = \text{grad}\,\dot{\mathbf{u}} = \dot{\mathbf{F}}\mathbf{F}^{-1}$, respectively. The first Piola stress in the base state is given by $\mathbf{S} = J\mathbf{T}\mathbf{F}^{-\top} = \partial\psi/\partial\mathbf{F}$, and the increments in the first Piola stress are obtained by linearizing this constitutive equation: $\dot{\mathbf{S}} = \mathbb{C}_{\text{R}}\dot{\mathbf{F}}$, where $\mathbb{C}_{\text{R}} = \partial^2\psi/\partial\mathbf{F}\partial\mathbf{F}$ are the material moduli, evaluated in the base state. The stress increments, pushed forward to the deformed body, are given by $\dot{\mathbf{T}} = J^{-1}\dot{\mathbf{S}}\mathbf{F}^{\top} = \mathbb{C}\dot{\mathbf{H}}$, where

$$\mathbb{C}_{ijkl}(\mathbf{y}) = J^{-1}F_{jn}F_{lq}\frac{\partial^2\psi}{\partial F_{in}\partial F_{kq}}, \tag{B.1}$$

are the spatial moduli, evaluated at the base state. The incremental form of the equations of motion are

$$\text{div}\,\dot{\mathbf{T}} = \text{div}\,(\mathbb{C}\dot{\mathbf{H}}) = \rho\frac{\partial^2\dot{\mathbf{u}}}{\partial t^2}, \tag{B.2}$$

where $\rho = J^{-1}\rho_{\text{R}}$ is the spatial mass density, ρ_{R} is the referential mass density, and $\partial(\bullet)/\partial t$ is the spatial time derivative. Straightforward application of the chain rule leads to the following expression for the spatial tangents:

$$\mathbb{C}_{ijkl} = \frac{1}{2J}\mathbb{D}_{ijmn}\mathbb{L}_{mnpq}\mathbb{B}_{pqkl} - T_{il}\delta_{jk} \tag{B.3}$$

with

$$\mathbb{D} = \frac{\partial\mathbf{T}_{\text{K}}}{\partial\mathbf{E}}, \quad \mathbb{L} = \frac{\partial\,\ln(\mathbf{B})}{\partial\mathbf{B}}, \quad \text{and} \quad \mathbb{B}_{ijkl} = \delta_{ik}B_{jl} + \delta_{jk}B_{il}, \tag{B.4}$$

where $\mathbf{T}_{\text{K}} = J\mathbf{T}$ is the Kirchhoff stress. For a free-energy function based on logarithmic strain invariants, $\tilde{\psi}(K_1, K_2, K_3)$, straightforward calculations involving the chain rule lead to the following expression for the constitutive contribution to the spatial tangents:

$$\begin{aligned}\mathbb{D}_{ijkl} = \frac{\partial T_{K,ij}}{\partial E_{kl}} = \frac{\partial^2\tilde{\psi}}{\partial K_1^2}\delta_{ij}\delta_{kl} + \left(\frac{\partial^2\tilde{\psi}}{\partial K_2^2} - \frac{\partial\tilde{\psi}}{\partial K_2}\frac{1}{K_2} - \frac{\partial\tilde{\psi}}{\partial K_3}\frac{3K_3}{K_2^2}\right)N_{ij}N_{kl} + \frac{\partial^2\tilde{\psi}}{\partial K_3^2}\frac{1}{K_2^2}Y_{ij}Y_{kl} \\ + \left(\frac{\partial^2\tilde{\psi}}{\partial K_1\partial K_2} - 2\sqrt{6}\frac{\partial\tilde{\psi}}{\partial K_3}\frac{1}{K_2^2}\right)\left(\delta_{ij}N_{kl} + N_{ij}\delta_{kl}\right) \\ + \frac{\partial^2\tilde{\psi}}{\partial K_1\partial K_3}\frac{1}{K_2}\left(\delta_{ij}Y_{kl} + Y_{ij}\delta_{kl}\right) \\ + \left(\frac{\partial^2\tilde{\psi}}{\partial K_2\partial K_3}\frac{1}{K_2} - 3\frac{\partial\tilde{\psi}}{\partial K_3}\frac{1}{K_2^2}\right)\left(N_{ij}Y_{kl} + Y_{ij}N_{kl}\right) \\ + \left(\frac{\partial\tilde{\psi}}{\partial K_2}\frac{1}{K_2} - \frac{\partial\tilde{\psi}}{\partial K_3}\frac{3K_3}{K_2^2}\right)\left(\mathbb{I}_{ijkl} - \frac{1}{3}\delta_{ij}\delta_{kl}\right) \\ + \frac{3\sqrt{6}}{2}\frac{\partial\tilde{\psi}}{\partial K_3}\frac{1}{K_2^2}\left(\delta_{ik}N_{jl} + \delta_{il}N_{jk} + N_{ik}\delta_{jl} + N_{il}\delta_{jk}\right). \end{aligned} \tag{B.5}$$

Then, we consider a homogeneous base state of deformation in an infinite domain – so that the base-state motion field $\boldsymbol{\chi}$ varies linearly in $\mathbf{x}$, and the base-state deformation gradient $\mathbf{F}$, spatial logarithmic strain tensor $\mathbf{E}$, and spatial moduli $\mathbb{C}$ are all spatially constant – and consider solutions for the incremental displacement field in the following form:

$$\dot{\mathbf{u}} = \hat{\mathbf{u}}\, g(k(\mathbf{n}\cdot\mathbf{y} - ct)), \tag{B.6}$$

where $k\mathbf{n}$ is a wave vector, k is the wave-vector magnitude, $\mathbf{n}$ is the unit vector along the direction of the wave vector, c is the wave speed, $\hat{\mathbf{u}}$ is the amplitude of the incremental displacement, and g is a continuous and sufficiently differentiable (but otherwise arbitrary) function. Upon substituting (B.6) into (B.2), we obtain the following eigenvalue problem: $\mathbf{Q}(\mathbf{n})\hat{\mathbf{u}} = \rho c^2\hat{\mathbf{u}}$ with eigenvalues ρc^2 and eigenvectors $\hat{\mathbf{u}}$, where $Q_{ik}(\mathbf{n}) = \mathbb{C}_{ijkl}n_j n_l$ is the acoustic tensor in the deformed configuration for a unit vector $\mathbf{n}$, which is utilized to assess stability in Section 5.1.

Appendix C. Digital image correlation parameters and strain validation

The digital image correlation details for typical cases of each experimental DIC configuration used during a given experiment are provided in Table C.1. Noise-floor resolution is defined as the standard deviation of the DIC displacement reconstructed from sequential, nominally zero-displacement images. Spatial resolution is estimated to be half of the minimum subset size (see Landauer et al., 2018).

To document the process of strain computation using a global plane fit, Fig. C.1(a) shows a fitting plane result with a comparison between the measured field and the noise floor for a typical experiment (specimen 2, experiment 2) on the low density foam for an incremental displacement between −0.261 and −0.266 global axial engineering strain. The zero-displacement noise floor for a single image in the pair, generated by adding Poisson noise (shot noise) and white Gaussian noise (0 mean, 0.75% variance, sensor noise) at a typical noise floor between deformed images (see Landauer et al., 2018), is of similar magnitude to the differences between the fitted plane and the experimental displacement measurement. For the same case, the spatial distribution of the locally-differentiated strain component – i.e., $e_{11}(x_1, x_2)$ – is presented in Fig. C.1(b) for both the zero-displacement noise and −0.266 global axial engineering strain cases. Strain was computed from cumulated displacements via an Optimal-9 differentiation kernel, per Bar-Kochba et al. (2015), and the virtual strain gauge size was 80 px (640 μm). The distribution of measured strain values is generally a single-peak Gaussian (see Fig. C.1(c)), revealing the underlying random nature of the spatial variance in strain (Wang and Cuitiño, 2002)

Table C1
Typical digital image correlation specification details for the experimental calibration and validation test cases. In most cases, values reported are from the high density, specimen 3, experiment 1 data and are typical.

Specification	Simple compression	Simple tension	Shear without pre-compression	Shear with pre-compression	Tension with holes
Technique	Single Camera q-factor Based Digital Image Correlation (Landauer et al., 2018)				
Pre-filtering	Gaussian, [3px, 0.2]				
Min. Subset Size [px]	16				
Min. Step Size [px]	8				
Correlation criterion	Iterative FFT-based ZNCC				
Interpolation	Bicubic splines				
Camera noise [%]	0.30	0.25	0.25	0.25	0.25
Image size [px^2]	1200 × 350	800 × 1900	1300 × 1200	950 × 1200	1100 × 2000
Measurement points	6560	23,750	24,375	17,813	34,375
Number of images	260 (0.033 Hz)	250 (0.040 Hz)	180 (0.033 Hz)	200 (0.033 Hz)	235 (0.143 Hz)
Pixel-to-μm conversion	1px = 10μm	1px=8μm	1px=8μm	1px=8μm	1px = 8μm
Resolution: spatial (est.)	8px = 80μm	8px = 64μm	8px = 64μm	8px = 64μm	8px = 64μm
Resolution: noise [px]	0.05	0.0075	0.046	0.033	0.05

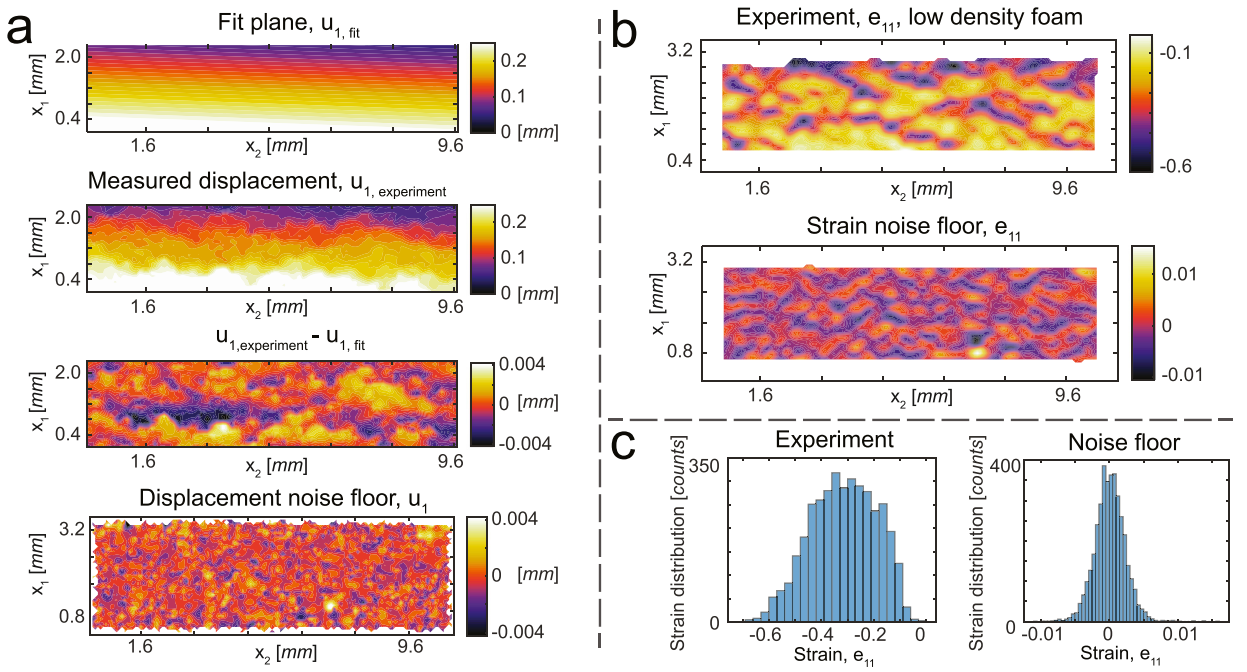


Fig. C.1. Spatial distribution of displacement and strain for a typical low density foam at −0.26 global axial engineering strain. (a) Incremental displacement from −0.261 to −0.266 global axial engineering strain with the zero-displacement noise profile, demonstrating the deviation from the best fit plane is of similar magnitude as the noise floor. (b) Comparison between experimental strain distribution at −0.26 global axial engineering strain to the noise floor for strain measurement for the image pair. (c) Histogram of experimental and noise floor strain values, showing a broad, slightly skewed Gaussian-like distribution for the experiment with mean centered about the applied global axial engineering strain and a zero-mean Gaussian-like profile for the noise floor.

References

Abaqus, 2018. Reference manuals. Dassault Systèmes Simulia Corp.

Anand, L., 1979. On H. Hencky's approximate strain-energy function for moderate deformations. J. Appl. Mech. 46, 78–82. doi:10.1115/1.3424532.

Anand, L., 1986. Moderate deformations in extension-torsion of incompressible isotropic elastic materials. J. Mech. Phys. Solids 34, 293–304. doi:10.1016/0022-5096(86)90021-9.

Anani, Y., Alizadeh, Y., 2011. Visco-hyperelastic constitutive law for modeling of foam's behavior. Mater. Des. 32, 2940–2948. doi:10.1016/j.matdes.2010.11.010.

Bar-Kochba, E., Toyjanova, J., Andrews, E., Kim, K., Franck, C., 2015. A fast iterative digital volume correlation algorithm for large deformations. Exper. Mech. 55, 261–274. doi:10.1007/s11340-014-9874-2.

Bergstrom, J., Boyce, M.C., 1998. Constitutive modeling of the large strain time-dependent behavior of elastomers. J. Mech. Phys. Solids 46, 931–954. doi:10.1016/S0022-5096(97)00075-6.

Bigoni, D., 2012. Nonlinear Solid Mechanics: Bifurcation Theory and Material Instability. Cambridge University Press.

Blatz, P.J., Ko, W.L., 1962. Application of finite elastic theory to the deformation of rubbery materials. Trans. Soc. Rheol. 6, 223–251. doi:10.1122/1.548937.

Briody, C., Duignan, B., Jerrams, S., Tiernan, J., 2012. The implementation of a visco-hyperelastic numerical material model for simulating the behaviour of polymer foam materials. Comput. Mater. Sci. 64, 47–51. doi:10.1016/j.commatsci.2012.04.012.

Brydon, A.D., Bardenhagen, S.G., Miller, E.A., Seidler, G.T., 2005. Simulation of the densification of real open-celled foam microstructures. J. Mech. Phys. Solids 53, 2638–2660. doi:10.1016/j.jmps.2005.07.007.

Chen, W., Lu, Z., Pan, B., Guo, J., Li, Y., 2012. Nonlinear behavior of bumper foams under uniaxial compressive cyclic loading. Mater. Design 35, 491–497. doi:10.1016/j.matdes.2011.09.042.

Chen, Y., Guo, W., Yang, P., Zhao, J., Guo, Z., Dong, L., Zhong, Z., 2018. Constitutive modeling of neo-Hookean materials with spherical voids in finite deformation. Extreme Mech. Lett. 24, 47–57. doi:10.1016/j.eml.2018.08.007.

Ciambella, J., Saccomandi, G., 2014. A continuum hyperelastic model for auxetic materials. Proc. R. Soc. Lond. A 470, 20130691. doi:10.1098/rspa.2013.0691.

Criscione, J.C., 2004. Rivlin's representation formula is ill-conceived for the determination of response functions via biaxial testing. In: The Rational Spirit in Modern Continuum Mechanics. Springer, pp. 197–215.

Criscione, J.C., Humphrey, J.D., Douglas, A.S., Hunter, W.C., 2000. An invariant basis for natural strain which yields orthogonal stress response terms in isotropic hyperelasticity. J. Mech. Phys. Solids 48, 2445–2465. doi:10.1016/S0022-5096(00)00023-5.

Dacorogna, B., 2001. Necessary and sufficient conditions for strong ellipticity of isotropic functions in any dimension. Discrete Cont. Dyn. Syst. Ser. B 1, 257–263. doi:10.3934/dcdsb.2001.1.257.

Danielsson, M., Parks, D.M., Boyce, M.C., 2004. Constitutive modeling of porous hyperelastic materials. Mech. Mater. 36, 347–358. doi:10.1016/S0167-6636(03)00064-4.

Deshpande, V.S., Fleck, N.A., 2001. Multi-axial yield behaviour of polymer foams. Acta Mater. 49, 1859–1866. doi:10.1016/S1359-6454(01)00058-1.

Diani, J., Gilormini, P., 2005. Combining the logarithmic strain and the full-network model for a better understanding of the hyperelastic behavior of rubber-like materials. J. Mech. Phys. Solids 53, 2579–2596. doi:10.1016/j.jmps.2005.04.011.

Elliott, J.A., Windle, A.H., Hobdell, J.R., Eeckhaut, G., Oldman, R.J., Ludwig, W., Boller, E., Cloetens, P., Baruchel, J., 2002. *In-situ* deformation of an open-cell flexible polyurethane foam characterised by 3D computed microtomography. J. Mater. Sci. 37, 1547–1555. doi:10.1023/a:1014920902712.

Gibson, L.J., Ashby, M.F., 1997. Cellular Solids: Structures and Properties, 2nd ed. Cambridge University Press.

Gong, L., Kyriakides, S., 2005. Compressive response of open cell foams. Part II: Initiation and evolution of crushing. Int. J. Solids Struct. 42, 1381–1399. doi:10.1016/j.ijsolstr.2004.07.024.
Gong, L., Kyriakides, S., Jang, W., 2005. Compressive response of open-cell foams. Part I: Morphology and elastic properties. Int. J. Solids Struct. 42, 1355–1379. doi:10.1016/j.ijsolstr.2004.07.023.
Guo, Z., Caner, F., Peng, X., Moran, B., 2008. On constitutive modelling of porous neo-Hookean composites. J. Mech. Phys. Solids 56, 2338–2357. doi:10.1016/j.jmps.2007.12.007.
Hencky, H., 1931. The law of elasticity for isotropic and quasi-isotropic substances by finite deformations. J. Rheol. 2, 169–176. doi:10.1122/1.2116361.
Hencky, H., 1933. The elastic behavior of vulcanised rubber. Rubber Chem. Technol. 6, 217–224. doi:10.5254/1.3547545.
Hill, R., 1962. Acceleration waves in solids. J. Mech. Phys. Solids 10, 1–16. doi:10.1016/0022-5096(62)90024-8.
Horgan, C.O., Murphy, J.G., 2009. A generalization of Hencky's strain-energy density to model the large deformations of slightly compressible solid rubbers. Mech. Mater. 41, 943–950. doi:10.1016/j.mechmat.2009.03.001.
Jemiolo, S., Turteltaub, S., 2000. Parametric model for a class of foam-like isotropic hyperelastic materials. J. Appl. Mech. 67, 248–254. doi:10.1115/1.1305277.
Ju, M.L., Mezghani, S., Jmal, H., Dupuis, R., Aubry, E., 2013. Parameter estimation of a hyperelastic constitutive model for the description of polyurethane foam in large deformation. Cellular Poly. 32, 21–40. doi:10.1177/026248931303200102.
Kakavas, P.A., 2000. A new development of the strain energy function for hyperelastic materials using a logarithmic strain approach. J. Appl. Pol. Sci. 77, 660–672. doi:10.1002/(SICI)1097-4628(20000718)77:3<660::AID-APP21>3.0.CO;2-A.
Lakes, R., Rosakis, P., Ruina, A., 1993. Microbuckling instability in elastomeric cellular solids. J. Mater. Sci. 28, 4667–4672. doi:10.1007/bf00414256.
Landauer, A.K., Patel, M., Henann, D.L., Franck, C., 2018. A q-factor-based digital image correlation algorithm (qDIC) for resolving finite deformations with degenerate speckle patterns. Exper. Mech. 58, 815–830. doi:10.1007/s11340-018-0377-4.
Liu, Q., Subhash, G., 2004. A phenomenological constitutive model for foams under large deformations. Pol. Eng. Sci. 44, 463–473. doi:10.1002/pen.20041.
Lopez-Pamies, O., Ponte Castañeda, P., 2007. Homogenization-based constitutive models for porous elastomers and implications for microscopic instabilities: I – analysis. J. Mech. Phys. Solids 55, 1677–1701. doi:10.1016/j.jmps.2007.01.007.
Lopez-Pamies, O., Ponte Castañeda, P., 2007. Homogenization-based constitutive models for porous elastomers and implications for microscopic instabilities: II – results. J. Mech. Phys. Solids 55, 1702–1728. doi:10.1016/j.jmps.2007.01.008.
Mills, N.J., Gilchrist, A., 2000. Modelling the indentation of low density polymer foams. Cellular Poly. 19, 389–412.
Murphy, J.G., 2000. Strain energy function for a poisson power law function in simple tension of compressible hyperelastic materials. J. Elast. 60, 151–164. doi:10.1023/A:1010843015909.
Neilsen, M.K., Krieg, R.D., Schreyer, H.L., 1995. A constitutive theory for rigid polyurethane foam. Poly. Eng. Sci. 35, 387–394. doi:10.1002/pen.760350503.
Ogden, R., 1984. Non-Linear Elastic Deformation. Dover Publications.
Ogden, R.W., 1972. Large deformation isotropic elasticity: On the correlation of theory and experiment for compressible rubberlike solids. Proc. R. Soc. Lond. Ser. A, Math. Phys. Sci. 328, 567–583. doi:10.1098/rspa.1972.0096.
Ogden, R.W., 1972. Large deformation isotropic elasticity: on the correlation of theory and experiment for incompressible rubberlike solids. Proc. R. Soc. Lond. Ser. A, Math. Phys. Sci. 326, 565–584. doi:10.1098/rspa.1972.0026.
Petre, M.T., Erdemir, A., Cavanagh, P.R., 2006. Determination of elastomeric foam parameters for simulations of complex loading. Comput. Methods Biomech. Biomed. Eng. 9, 231–242. doi:10.1080/10255840600747620.
Reese, S., Govindjee, S., 1998. A theory of finite viscoelasticity and numerical aspects. Int. J. Solids Struct. 35, 3455–3482. doi:10.1016/S0020-7683(97)00217-5.
Sabuwala, T., Gioia, G., 2013. Skeleton-and-bubble model of polyether-polyurethane elastic open-cell foams for finite element analysis at large deformations. J. Mech. Phys. Solids 61, 886–911. doi:10.1016/j.jmps.2012.10.007.
Shrimali, B., Lefèvre, V., Lopez-Pamies, O., 2019. A simple explicit homogenization solution for the macroscopic elastic response of isotropic porous elastomers. J. Mech. Phys. Solids 122, 364–380. doi:10.1016/j.jmps.2018.09.026.
Shulmeister, V., van der Burg, M.W.D., van der Giessen, E., Marissen, R., 1998. A numerical study of large deformations of low-density elastomeric open-cell foams. Mech. Mater. 30, 125–140. doi:10.1016/S0167-6636(98)00033-7.
Storakers, B., 1986. On material representation and constitutive branching in finite compressible elasticity. J. Mech. Phys. Solids 34, 125–145. doi:10.1016/0022-5096(86)90033-5.
Tang, M., Huang, G., Zhang, H., Liu, Y., Chang, H., Song, H., Xu, D., Wang, Z., 2017. Dependences of rheological and compression mechanical properties on cellular structures for impact-protective materials. ACS Omega 2, 2214–2223. doi:10.1021/acsomega.7b00242.
Thomas, T.Y., 1961. Plastic Flow and Fracture in Solids. Academic Press.
Triantafyllidis, N., Aifantis, E., 1986. A gradient approach to localization of deformation. I. Hyperelastic materials. J. Elast. 16, 225–237. doi:10.1007/BF00040814.
Wang, S., Chester, S., 2018. Experimental characterization and continuum modeling of inelasticity in filled rubber-like materials. Int. J. Solids Struct. 136–137, 125–136. doi:10.1016/j.ijsolstr.2017.12.010.
Wang, Y., Cuitiño, A.M., 2000. Three-dimensional nonlinear open-cell foams with large deformations. J. Mech. Phys. Solids 48, 961–988. doi:10.1016/S0022-5096(99)00060-5.
Wang, Y., Cuitiño, A.M., 2002. Full-field measurements of heterogeneous deformation patterns on polymeric foams using digital image correlation. Int. J. Solids Struct. 39, 3777–3796. doi:10.1016/S0020-7683(02)00176-2.
Wang, Y., Gioia, G., Cuitiño, A.M., 2000. The deformation habits of compressed open-cell solid foams. J. Eng. Mater. Technol. 122, 376–378. doi:10.1115/1.1288923.
Widdle Jr., R.D., Bajaj, A.K., Davies, P., 2008. Measurement of the Poisson's ratio of flexible polyurethane foam and its influence on a uniaxial compression model. Int. J. Eng. Sci. 46, 31–49. doi:10.1016/j.ijengsci.2007.09.002.
Xiao, H., 2013. An explicit, direct approach to obtain multi-axial elastic potentials which accurately match data of four benchmark tests for rubbery materials – Part 2: General deformations. Acta Mech. 224, 479–498. doi:10.1007/s00707-012-0768-z.
Yang, L.M., Shim, V.P.W., 2004. A visco-hyperelastic constitutive description of elastomeric foam. Int. J. Impact Eng. 30, 1099–1110. doi:10.1016/j.ijimpeng.2004.03.011.
Zhang, J., Lin, Z., Wong, A., Kikuchi, N., Li, V.C., Yee, A.F., 1997. Constitutive modeling and material characterization of polymeric foams. J. Eng. Mater. Technol. 119, 284–291. doi:10.1115/1.2812258.